

DUBAI
TECHBOOTS
LEAGUE
SumoX-26



Sumo Robot Training

Presented by:



Pool Prizes



PARTNERS & SPONSORS

POWERED BY INDUSTRY LEADERS



Qualcomm



CANADIAN
UNIVERSITY DUBAI
Your portal to Canadian education



POWERED BY

ARE YOU READY TO DOMINATE?

Join the Dubai Techbots League SumoX-26. Build your bot.
Code your strategy. Fight for the ring.

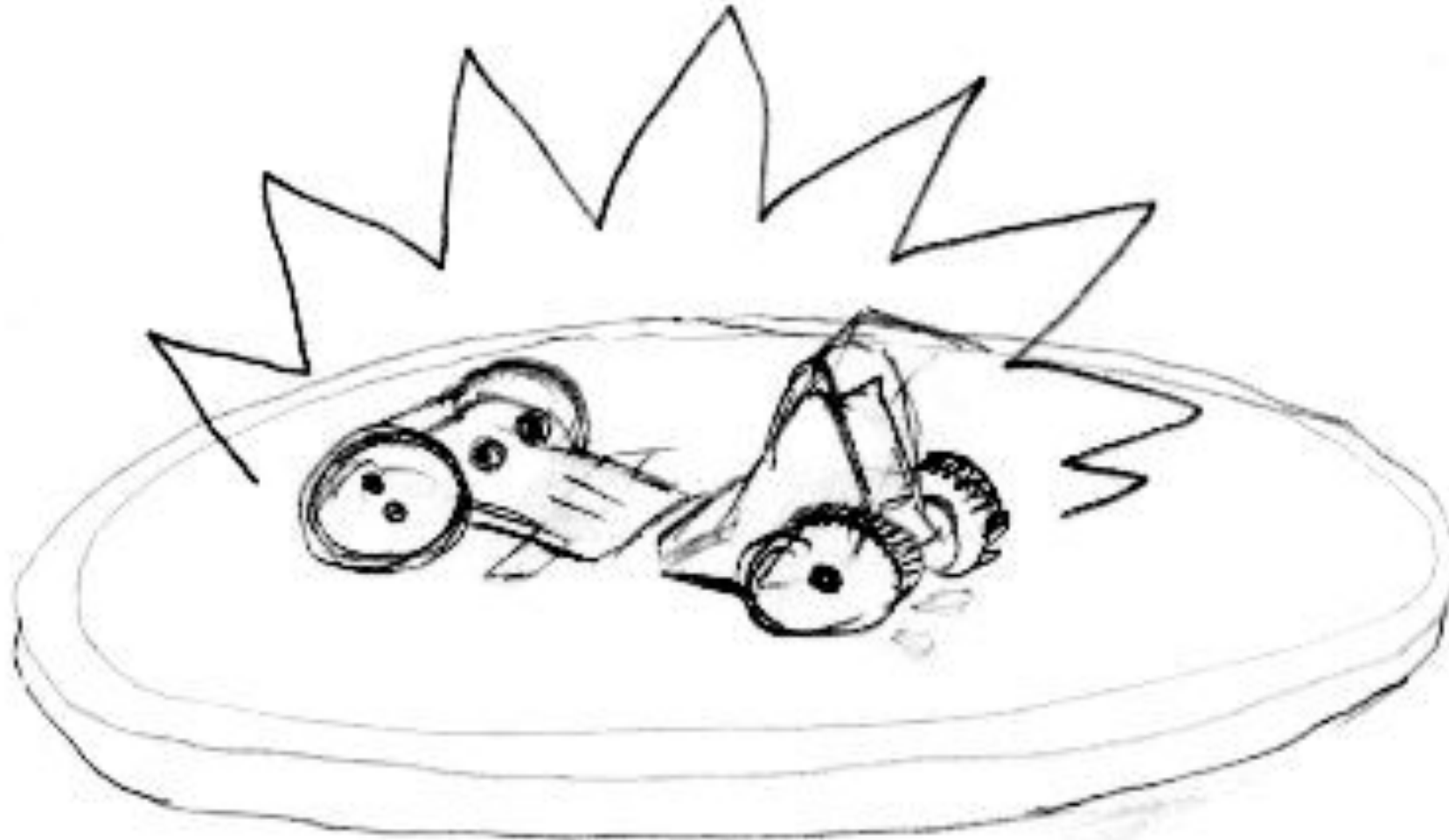


dubaitechbotsleague@besomi.com



**LET'S PLAY THE WOOL BALL GAME
AND THEN TEAM UP!**

WHAT IS SUMO ROBOTS?



WHAT IS SUMO ROBOTS?

Robot Control: Autonomous and Remote-control

Engineering Principles: Torque, mechanical engineering, electrical engineering, computer science (for autonomous classes.)

Event Summary:

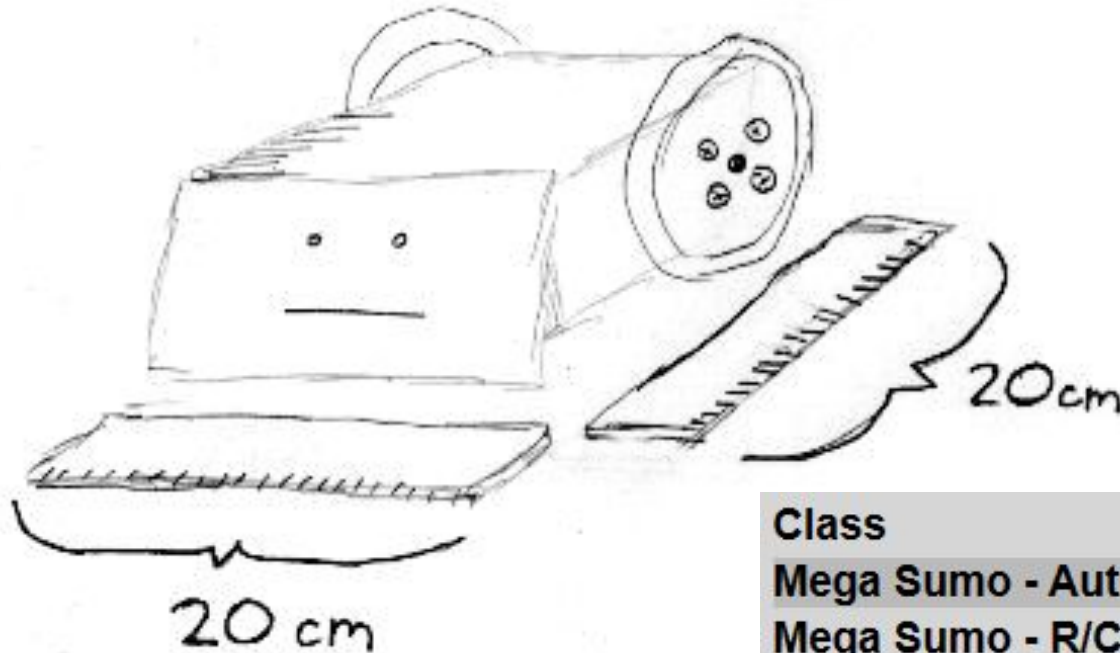
Two robots compete in a head-to-head match following the basic system of traditional human sumo matches. Robots are allowed no weapons, and are not allowed to flip each other. The sole purpose is a pushing match between the two robots to force the other from the arena. Multiple weight classes and control systems are allowed (autonomous compete against autonomous and R/C against R/C - they are separate classes and do not compete against each other.)

WHAT IS SUMO ROBOTS?

Dohyo Construction & Painting				Shikiri Lines			White Border
Class	Height	Diameter	Material	Width	Length	Separation	Width
Mega/Humanoid	5.00 cm	154.0 cm	Polyurethane coated steel plate	2 cm	20 cm	20 cm	5 cm
Mini/Kit	2.50 cm	77.0 cm	Laminate/Melamine coated wood	1 cm	10 cm	10 cm	2.5 cm
Micro	1.25 cm	38.5 cm	Melamine composite	0.5 cm	5 cm	5 cm	1.25 cm
Nano	0.625 cm	19.25 cm	Melamine composite	0.25 cm	2.5 cm	2.5 cm	0.625 cm



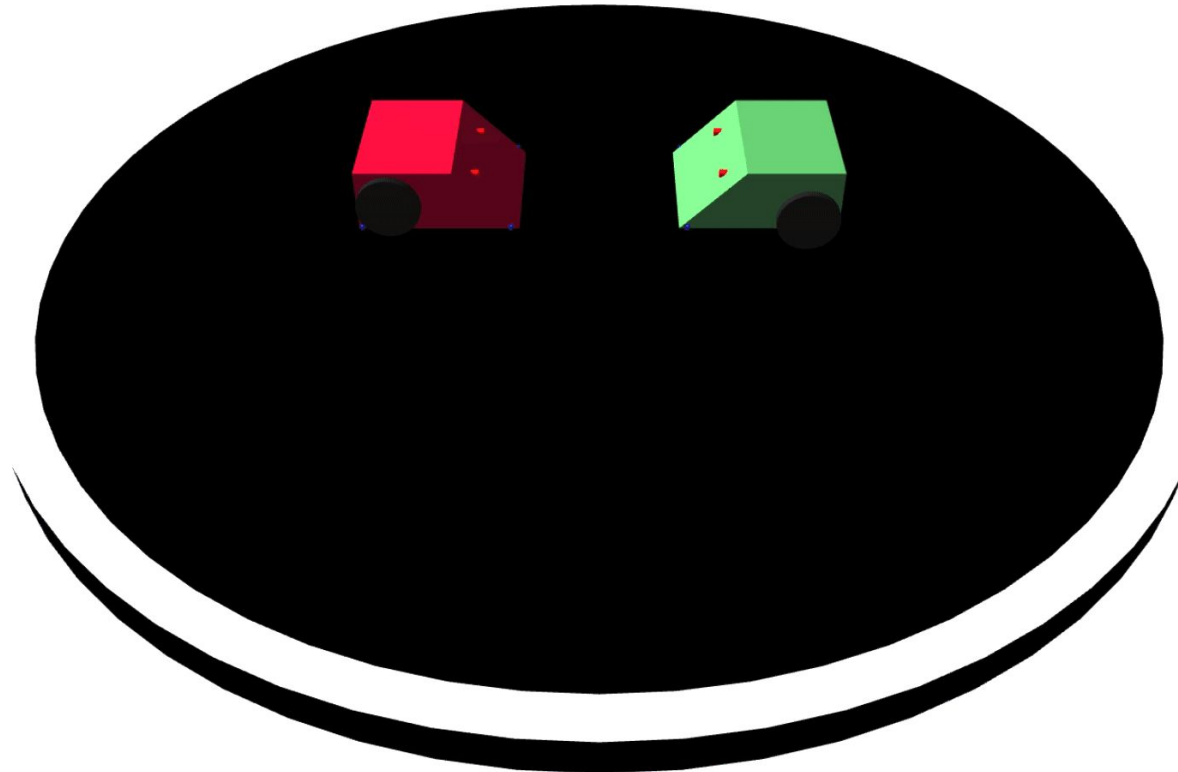
WHAT IS SUMO ROBOTS?



Class	Height	Width	Length	Weight
Mega Sumo - Auton	unlimited	20.0 cm	20.0 cm	3,000 g
Mega Sumo - R/C	unlimited	20.0 cm	20.0 cm	3,000 g
Mega Sumo - Network	unlimited	20.0 cm	20.0 cm	3,000 g
Humanoid - R/C	50 cm	20.0 cm	20.0 cm	4,000 g
Kit Sumo	unlimited	15.0 cm	15.0 cm	1,000 g
Mini Sumo	unlimited	10.0 cm	10.0 cm	500 g
Micro Sumo	5 cm	5.0 cm	5.0 cm	100 g
Nano Sumo	2.5 cm	2.5 cm	2.5 cm	25 g

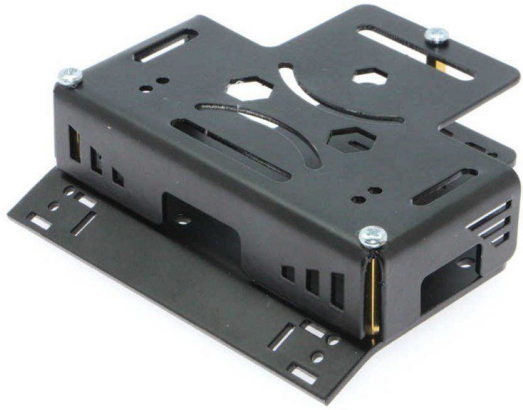
WHAT IS OUR GOAL?

To build an **autonomous robot** that **finds opponents** trying to fight and **push them** out of the ring, and **keep your robot inside the ring.**

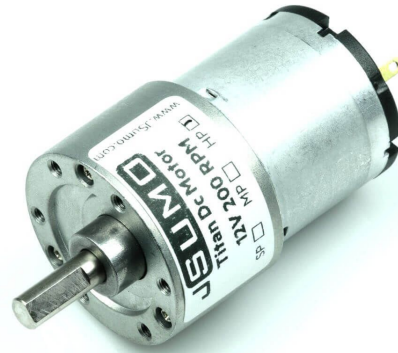


What a Sumo Robot Needs?

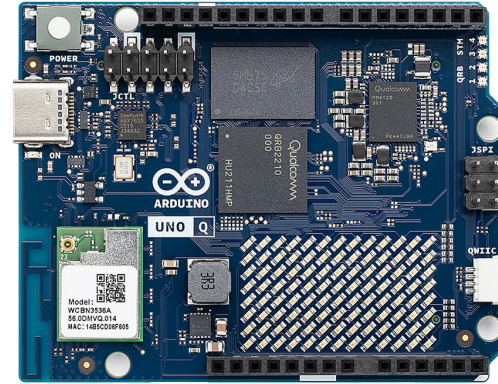
What a Sumo Robot Needs?



Chassis



Motors



Microcontroller



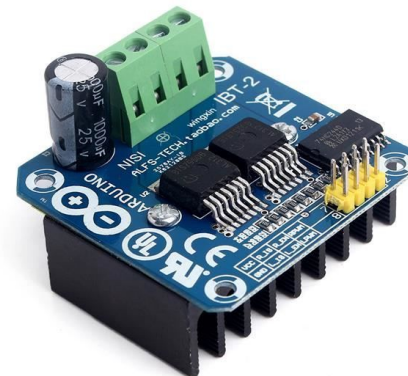
Line Sensors



Battery



Wheels

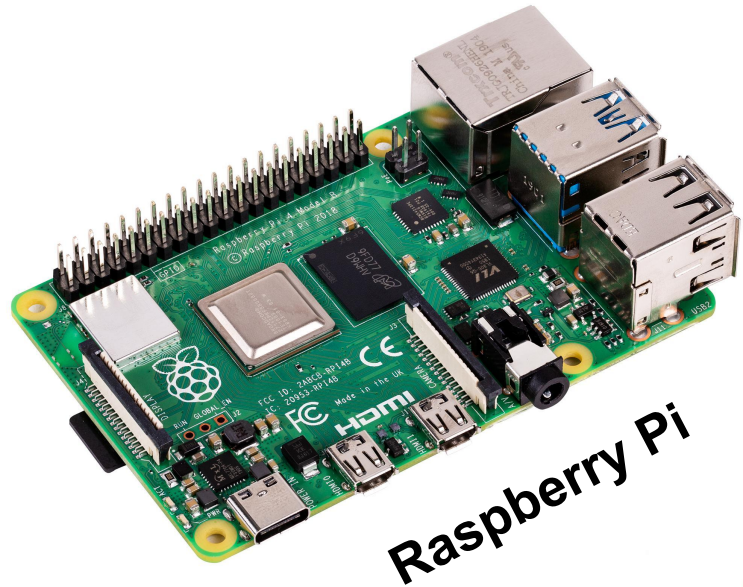


Motor Drivers

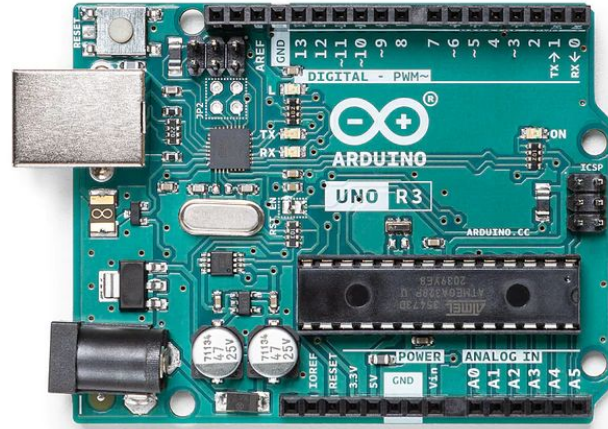


Obstacle Sensor

WHAT BRAIN TO CHOOSE?



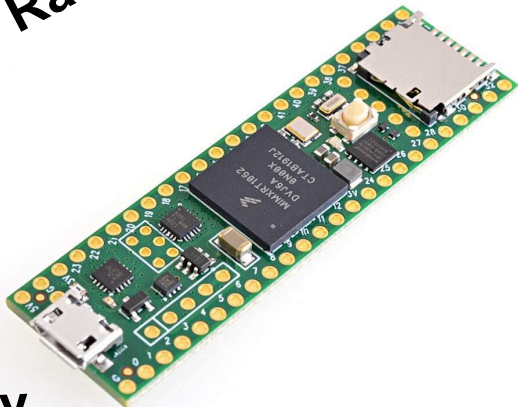
Raspberry Pi



Arduino



ESP32



Teensy



Ev3

TYPES OF ARDUINO BOARDS



ARDUINO UNO



ARDUINO LEONARDO



ARDUINO 101



LILYPAD ARDUINO USB



ARDUINO ROBOT



ARDUINO ESPLORA



ARDUINO MICRO



ARDUINO MEGA 2560



ARDUINO NANO



ARDUINO MINI



ARDUINO YÚN



ARDUINO ZERO

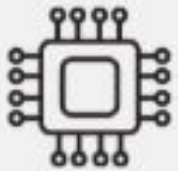
OUR CHOICE



Arduino UNO Q

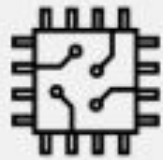
How?

The Arduino UNO Q combines on a single compact board:



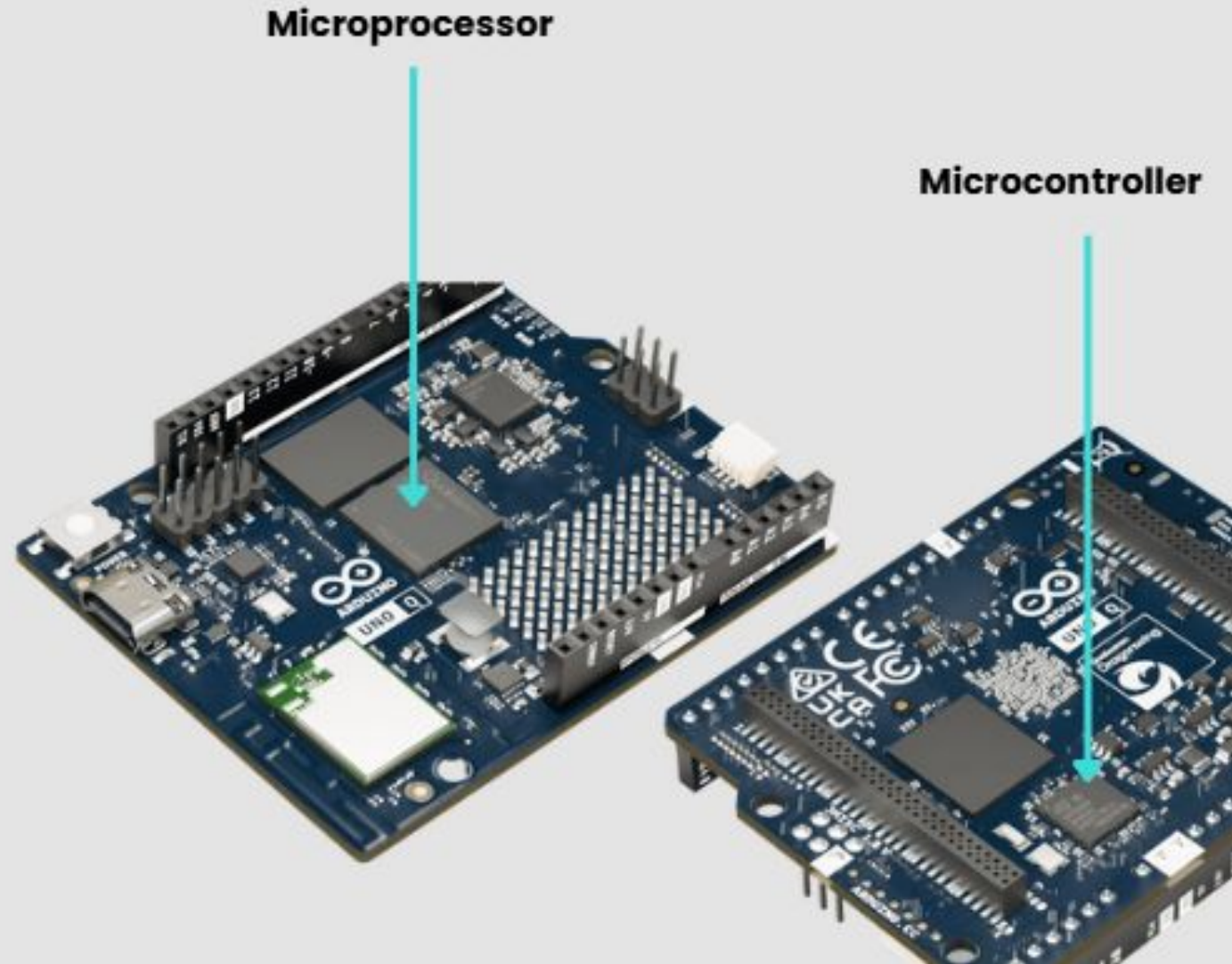
Real-time
microcontroller

STM32U585

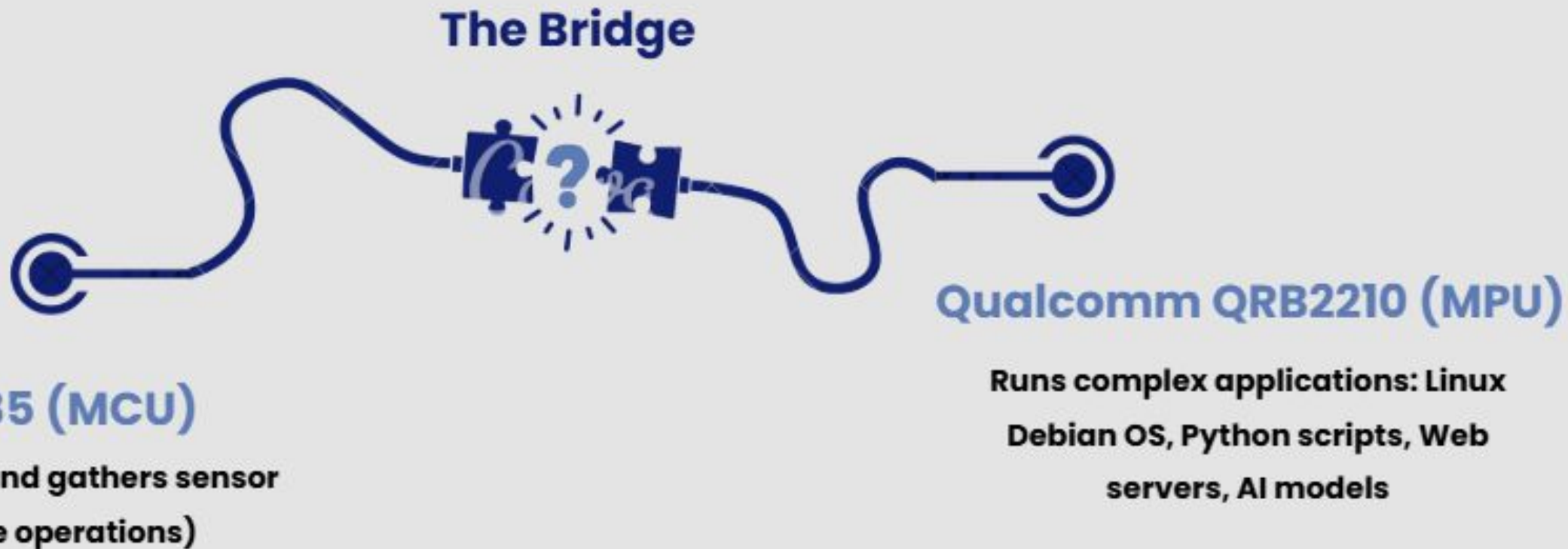


Powerful
microprocessor
running Linux

Qualcomm® QRB2210



The Arduino Bridge: MCU-MPU Communication



Programming Modes for Arduino UNO Q

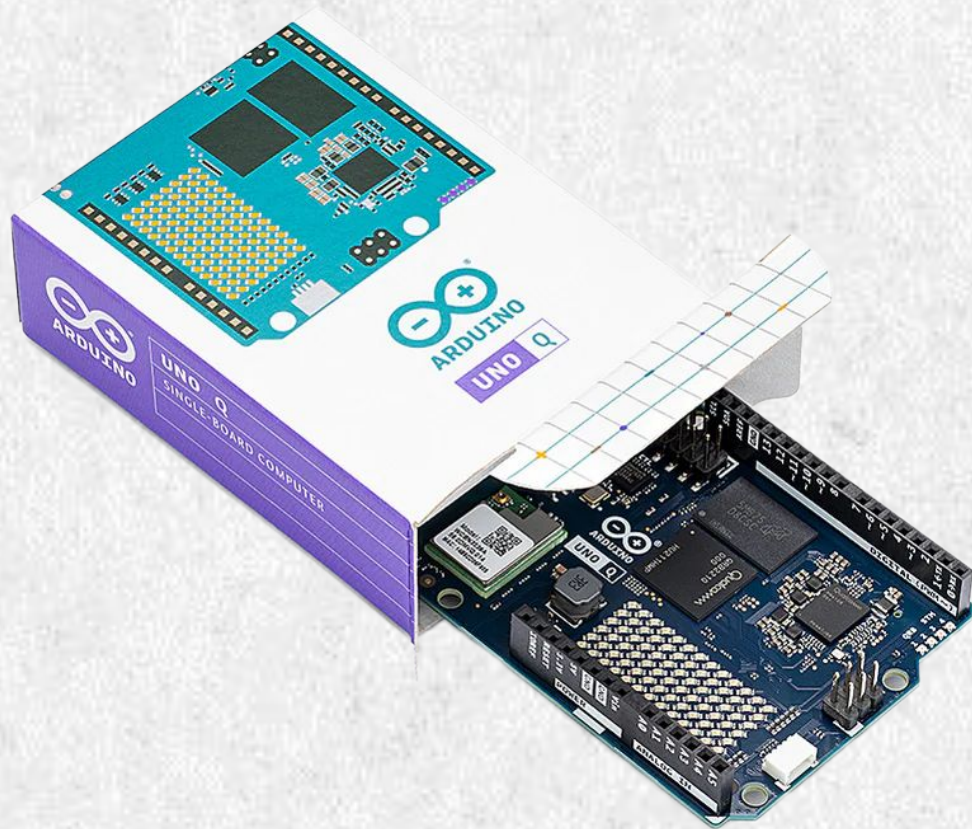
Two Flexible Development Approaches

PC-Hosted Mode



Standalone Mode



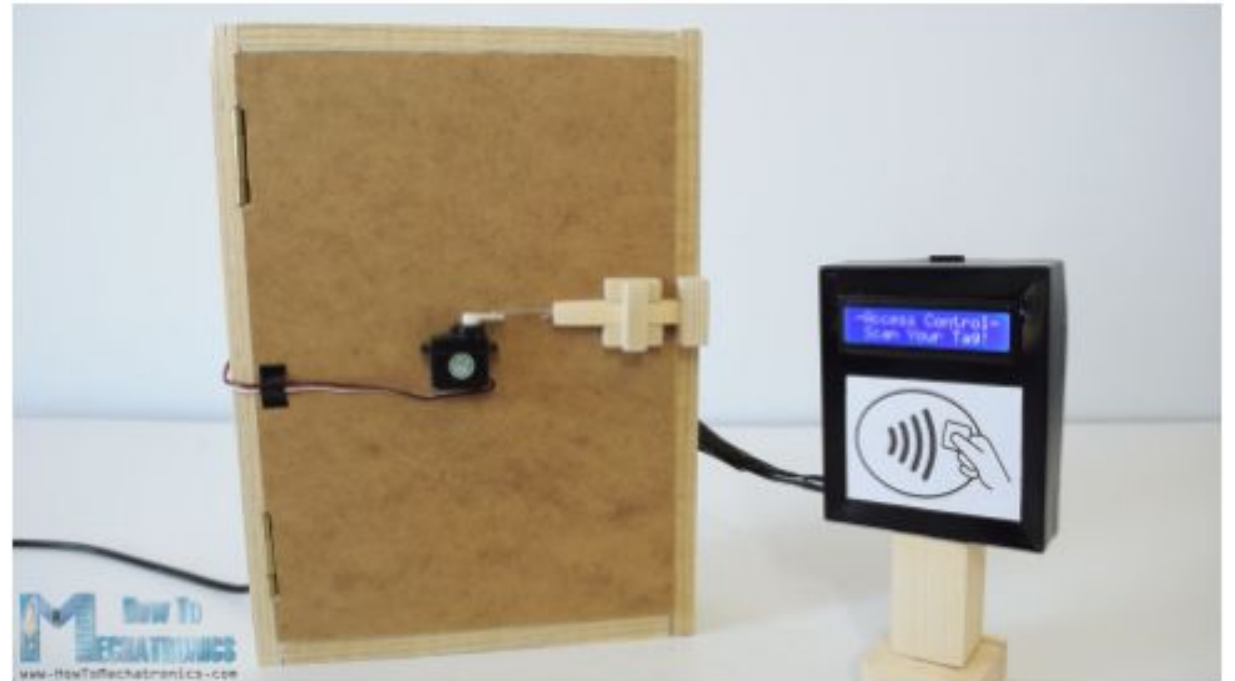
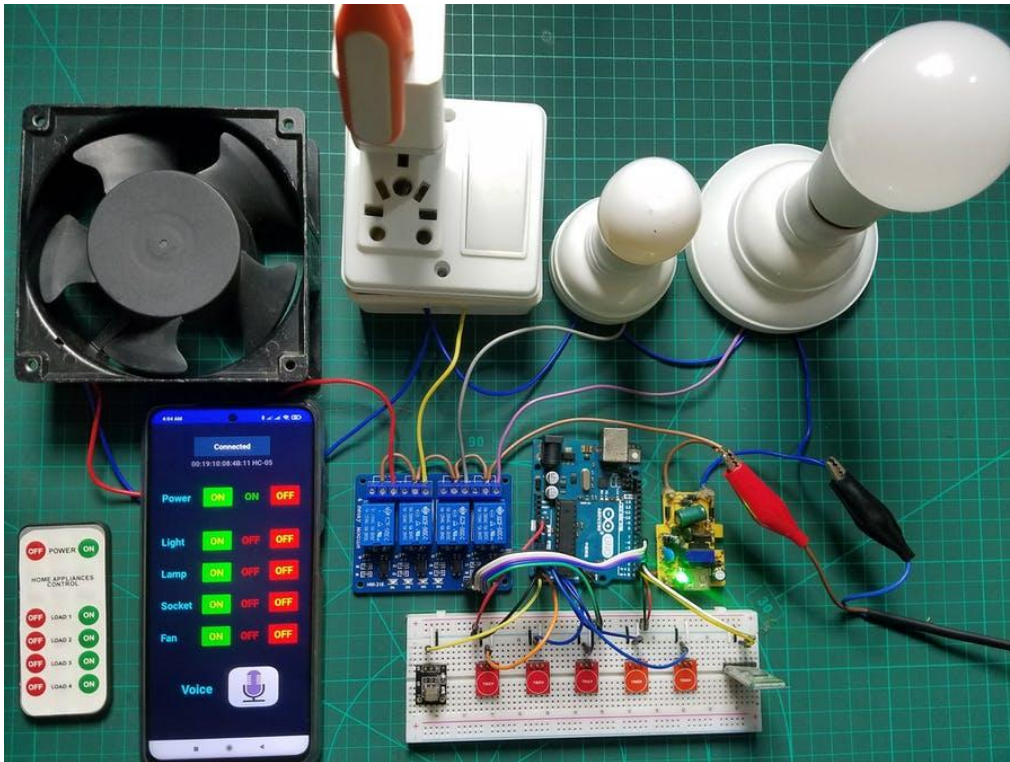


WHAT IS ARDUINO?

Arduino is an open-source electronics platform based on easy-to-use hardware and software.

APPLICATIONS OF ARDUINO

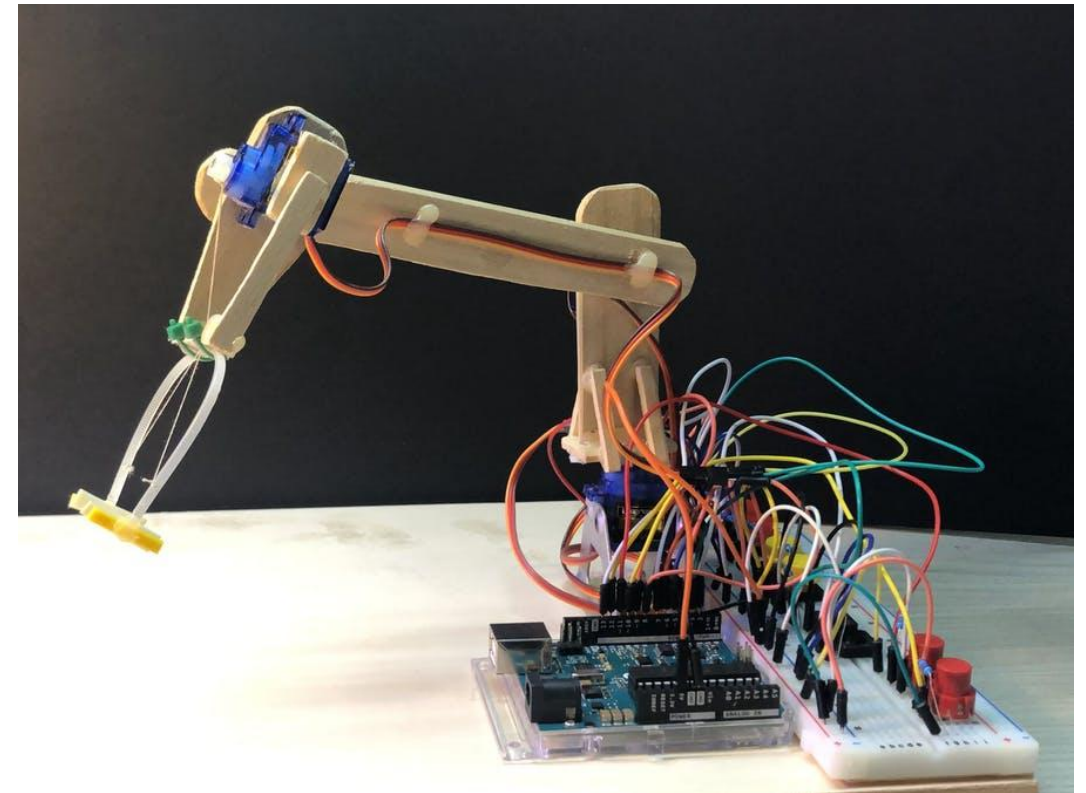
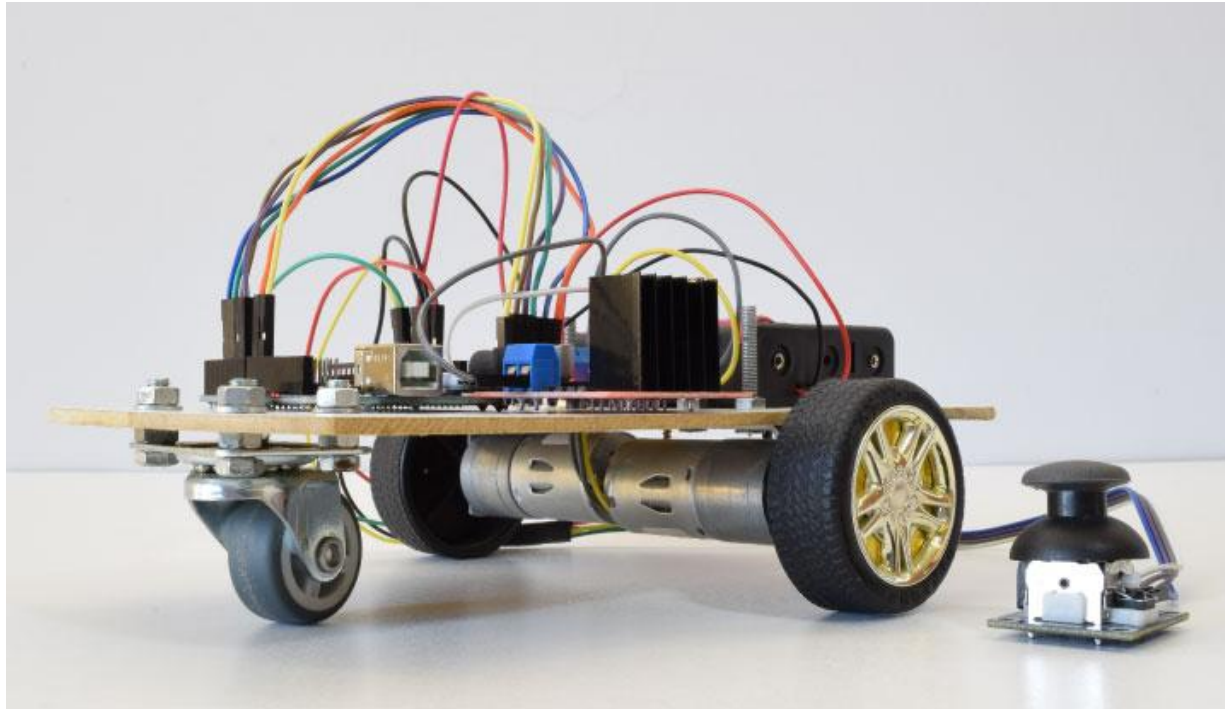
Real-world monitoring



Ref: [Howtomechanics](http://www.howtomechanics.com)

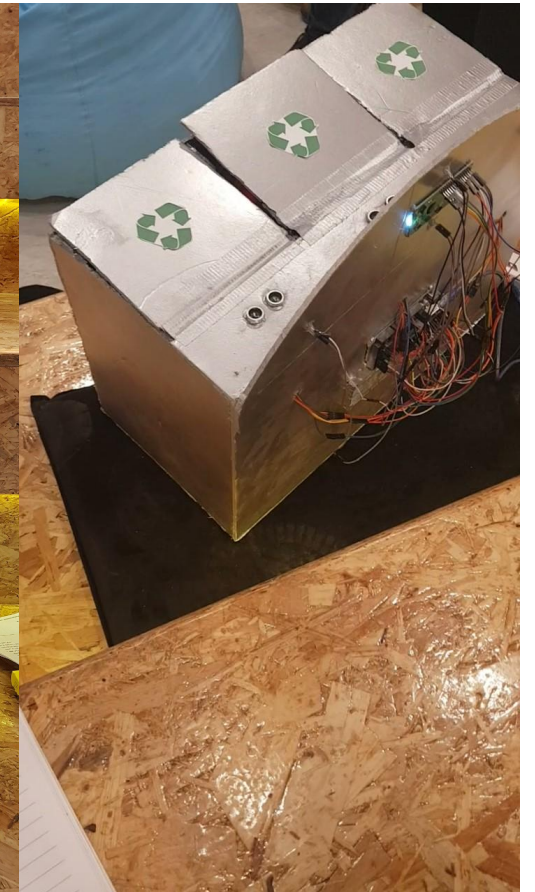
APPLICATIONS OF ARDUINO

Small-scale control



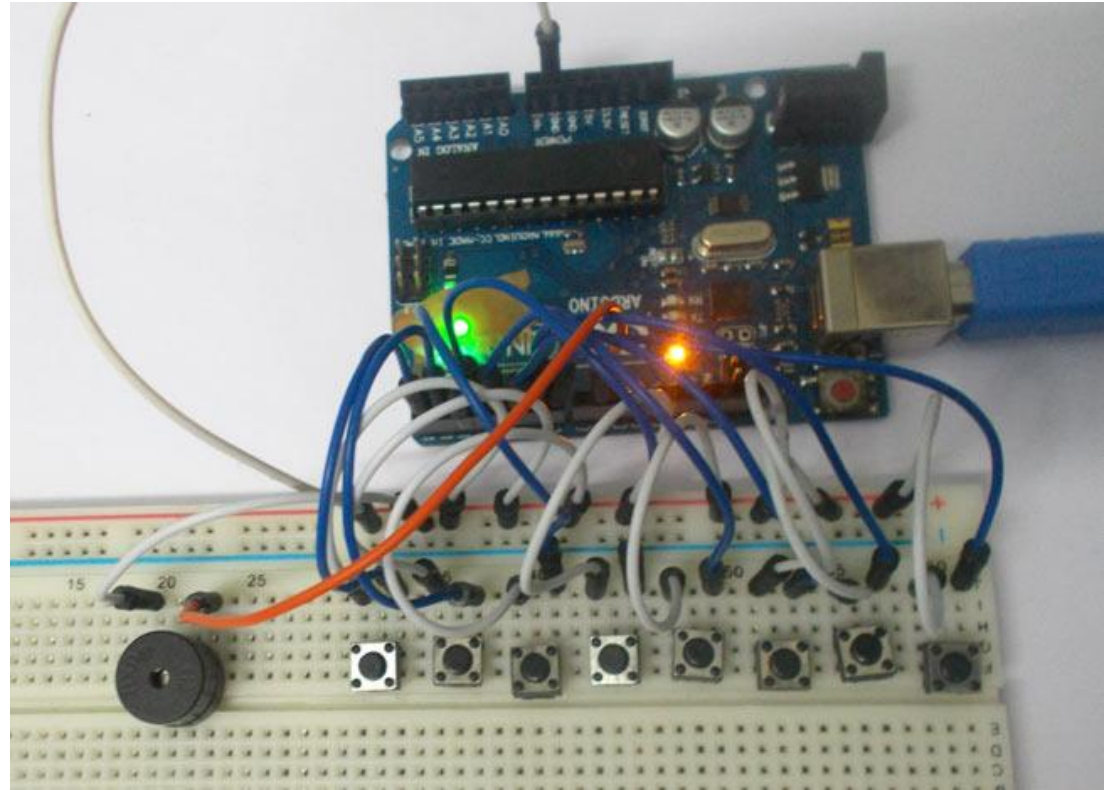
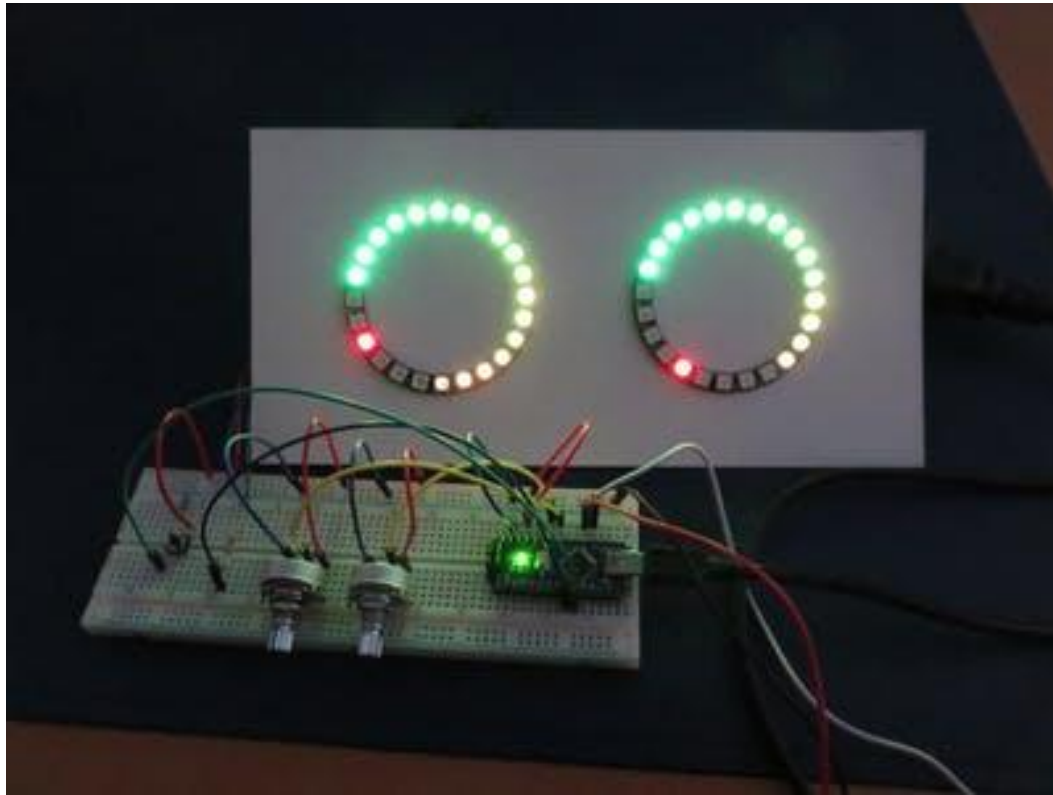
APPLICATIONS OF ARDUINO

Small-scale automation



APPLICATIONS OF ARDUINO

Performance art



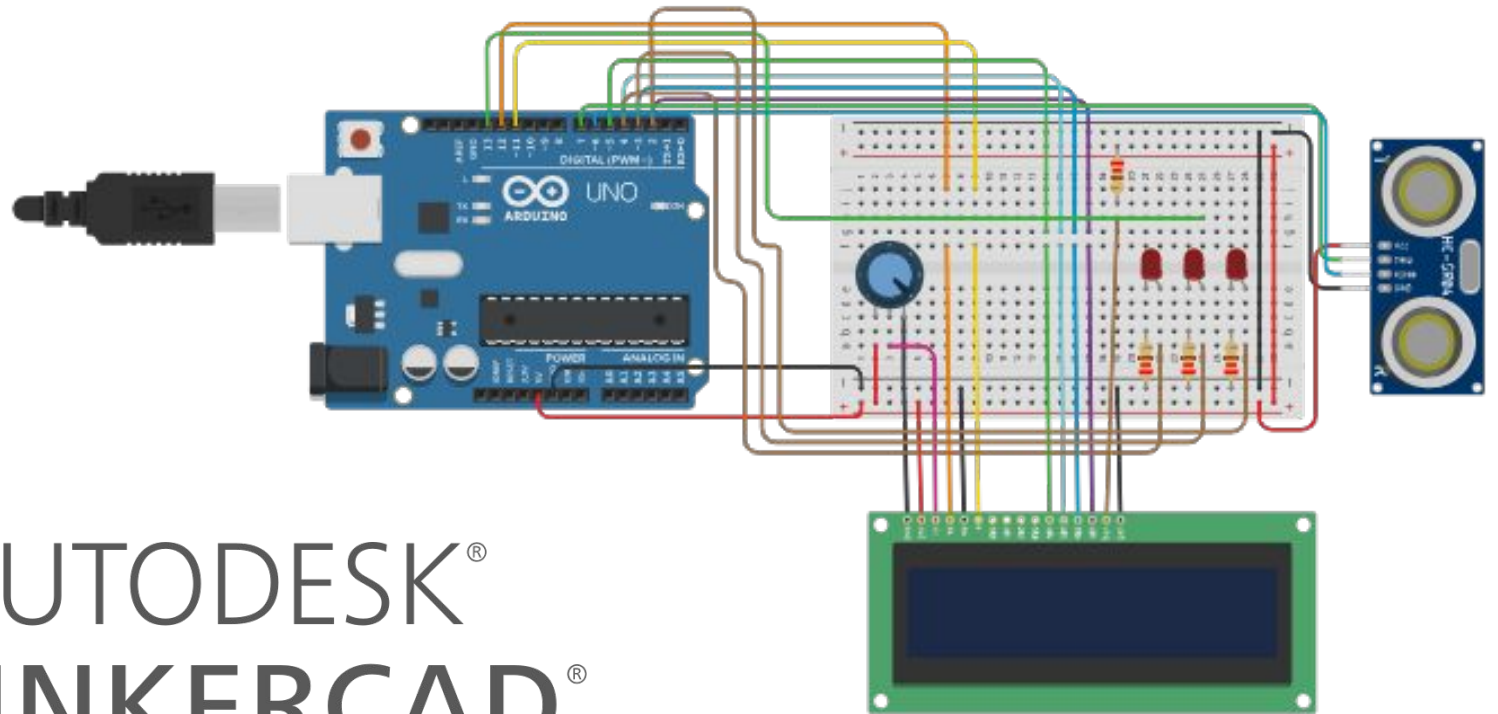
TWO WAYS TO PRACTICE WITH US

1

By simulation



AUTODESK®
TINKERCAD®



TWO WAYS TO PRACTICE WITH US

2

Physically by hands



CODING USING ARDUINO IDE



The screenshot shows the Arduino IDE window titled "Blink | Arduino 1.8.5". The interface includes a toolbar with icons for checking, uploading, and downloading code. A tab labeled "Blink §" is active. The main text area contains the following code:

```
This example code is in the public domain.

http://www.arduino.cc/en/Tutorial/Blink
*/

// the setup function runs once when you press reset or power the board
void setup() {
  // initialize digital pin LED_BUILTIN as an output.
  pinMode(LED_BUILTIN, OUTPUT);
}

// the loop function runs over and over again forever
void loop() {$
  digitalWrite(LED_BUILTIN, HIGH); // turn the LED on (HIGH is the voltage level)
  delay(1000); // wait for a second
  digitalWrite(LED_BUILTIN, LOW); // turn the LED off by making the voltage LOW
  delay(1000); // wait for a second
}
```

The status bar at the bottom indicates "32" and "Arduino/Genuino Uno on COM1".

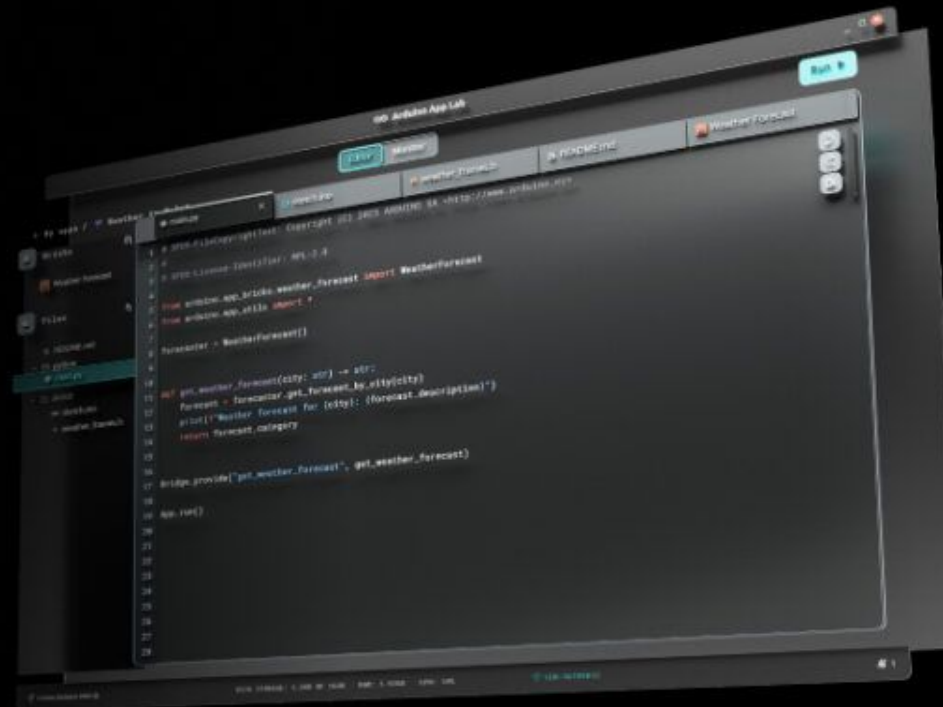


Programming Language



Arduino C

Arduino App Lab



ARDUINO
APP LAB



The Arduino App Lab



Unified Development Experience

- Combines Arduino Sketches, Python scripts, and AI models
- All managed from a single interface
- Preloaded on the UNO Q



Ready-to-Use Apps and Bricks

- Get started fast with example Apps with everything included
- Add plug-and-play features with pre-built Bricks

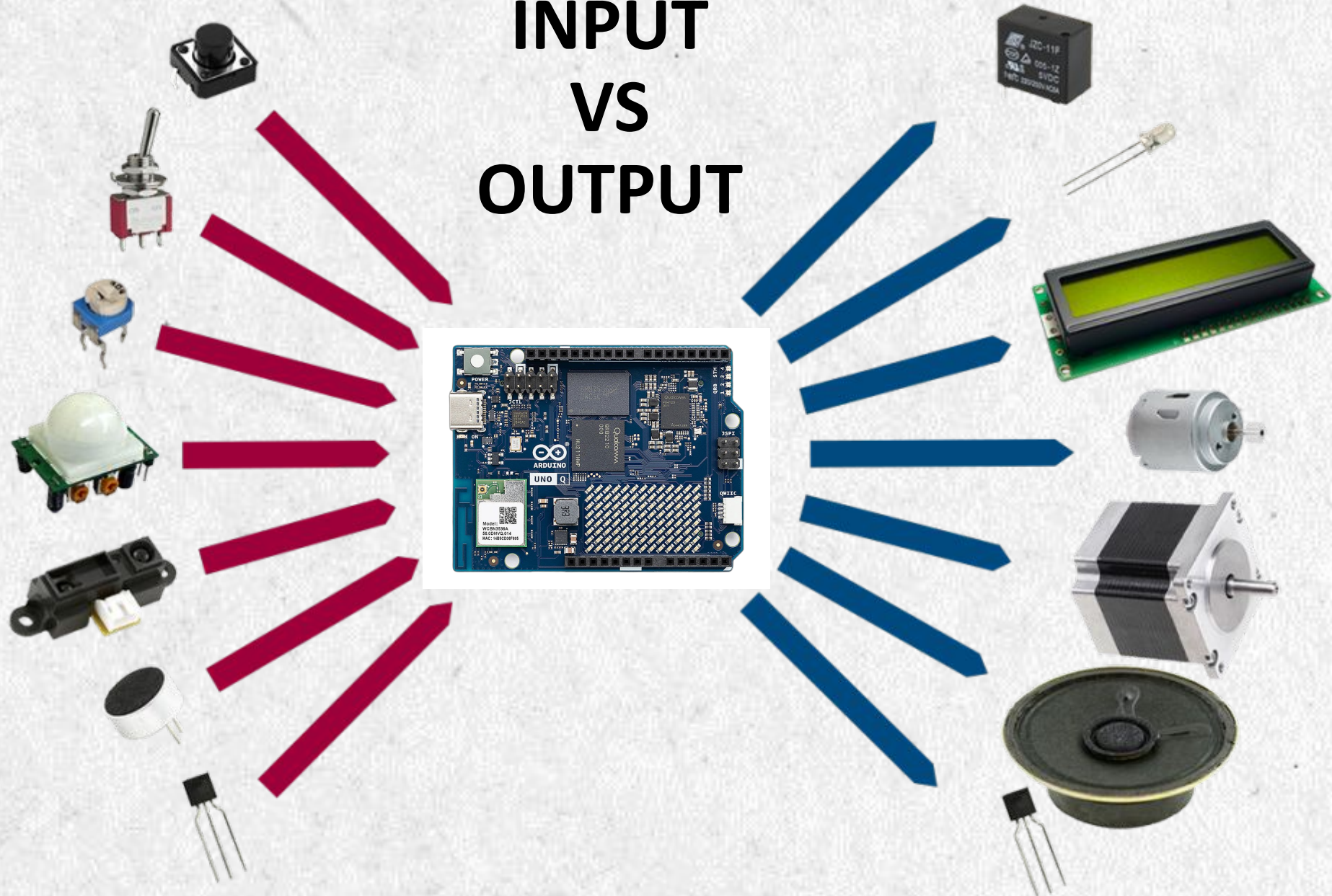


Pre-loaded AI Models

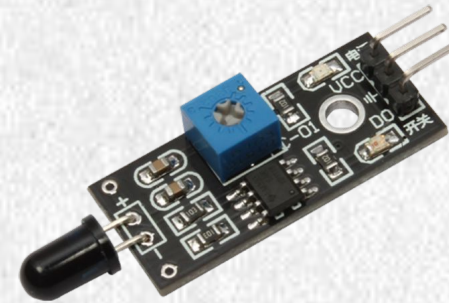
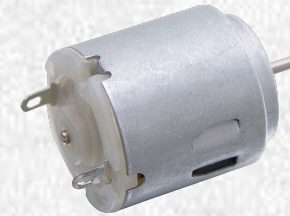
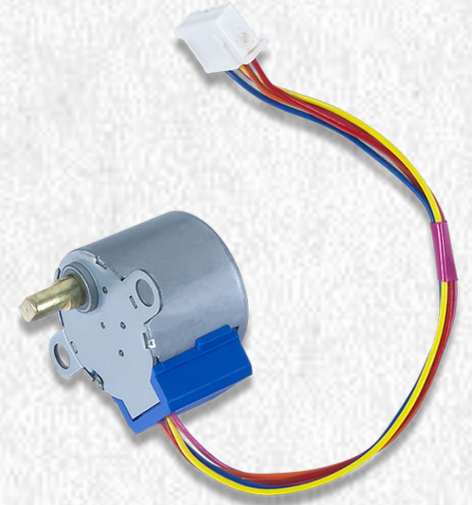
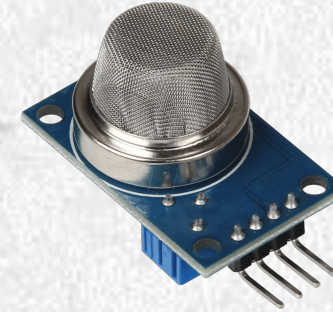
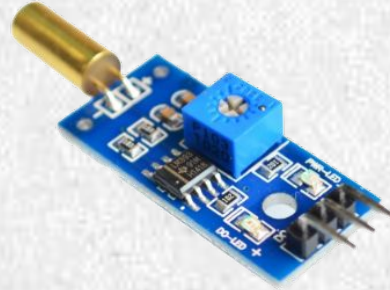
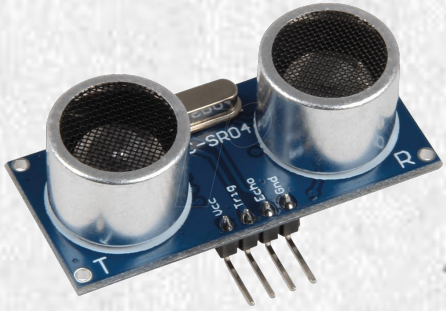
- Object/human detection
- Anomaly detection
- Image classification
- Sound recognition
- Keyword spotting

Inputs vs Outputs

INPUT VS OUTPUT

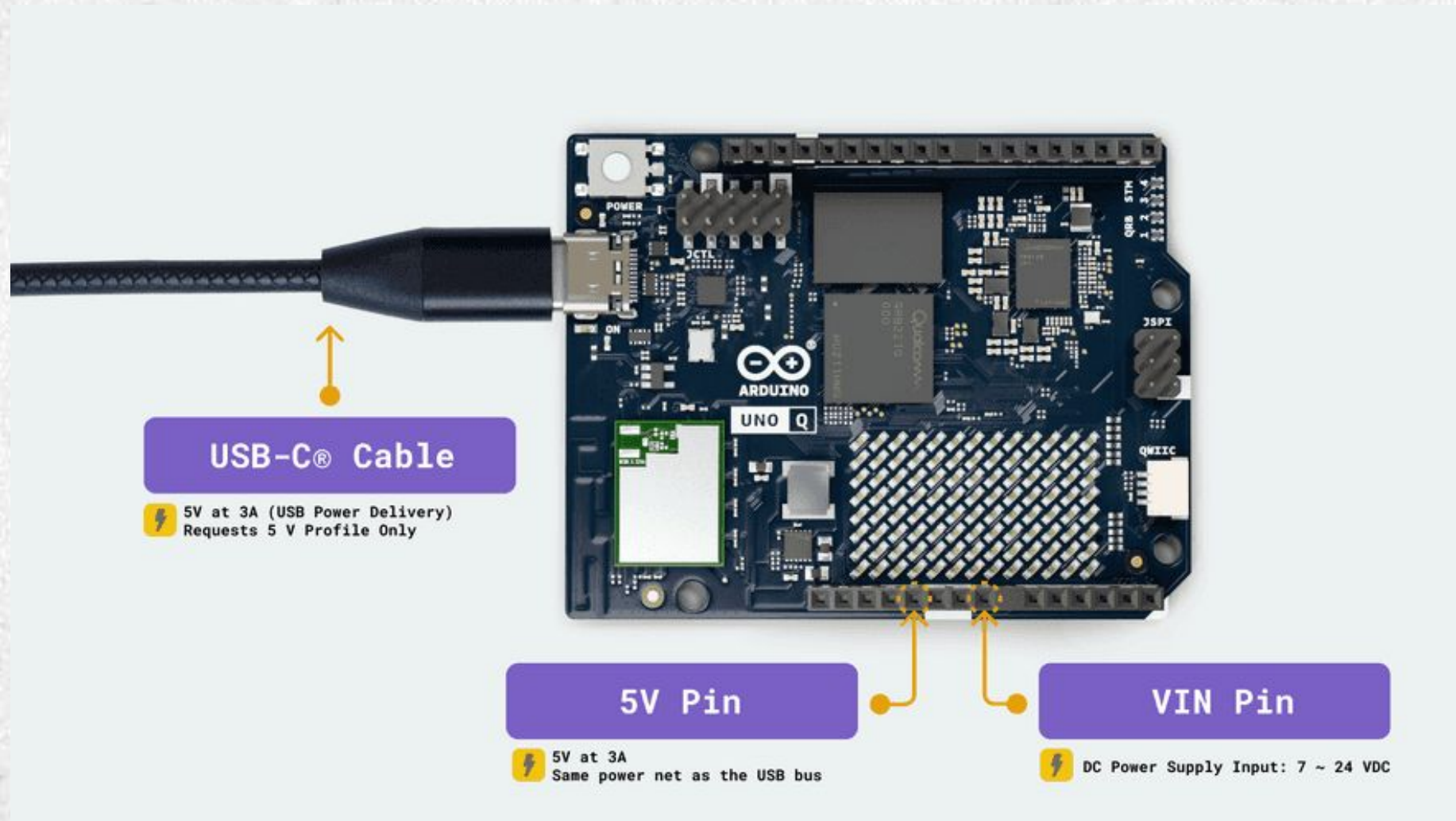


INPUT OR OUTPUT?



LET'S GET DEEPER

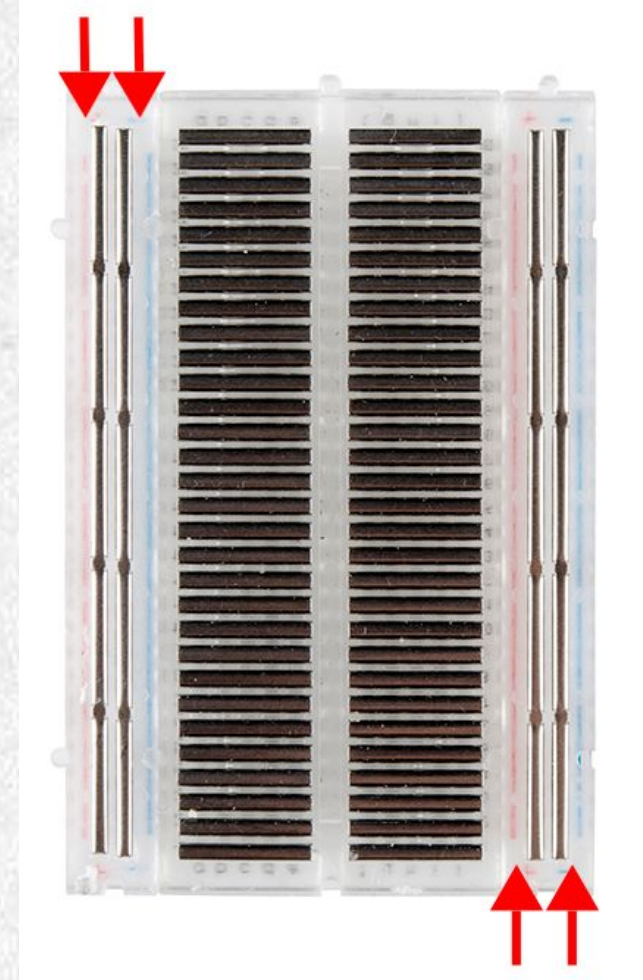
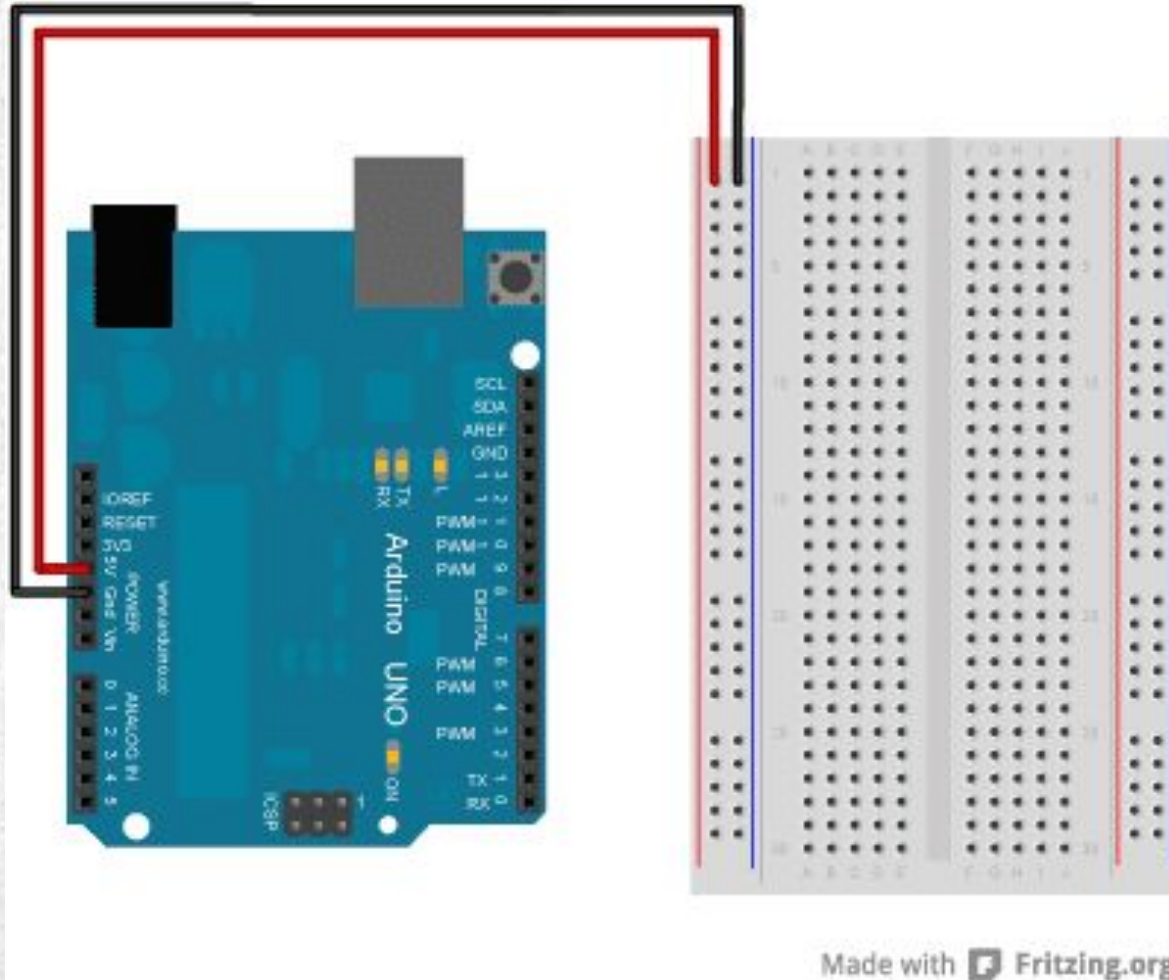
3 WAYS TO POWER THE ARDUINO BOARD





FEW THINGS WE SHOULD KNOW ABOUT

THE BREADBOARD



JUMPER WIRES



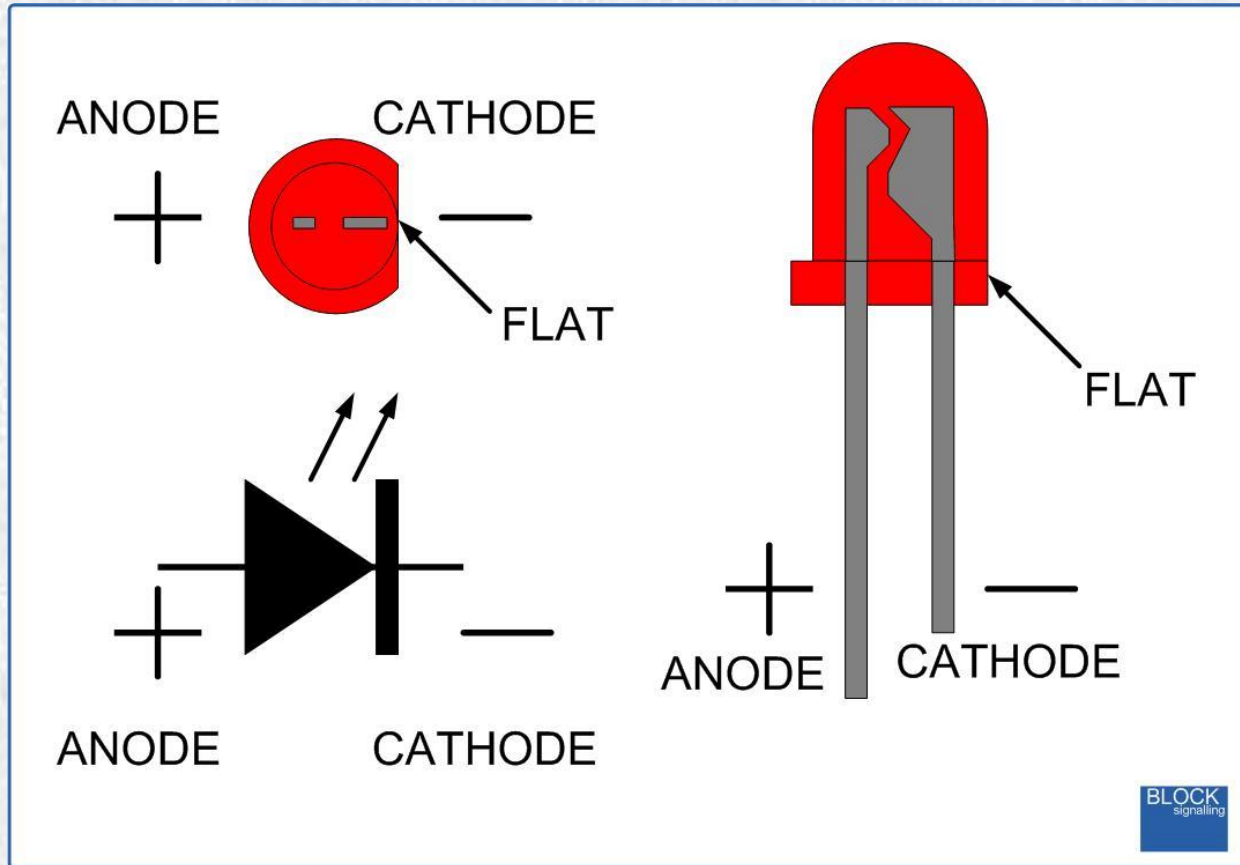
Male-to-Male

Male-to-Female

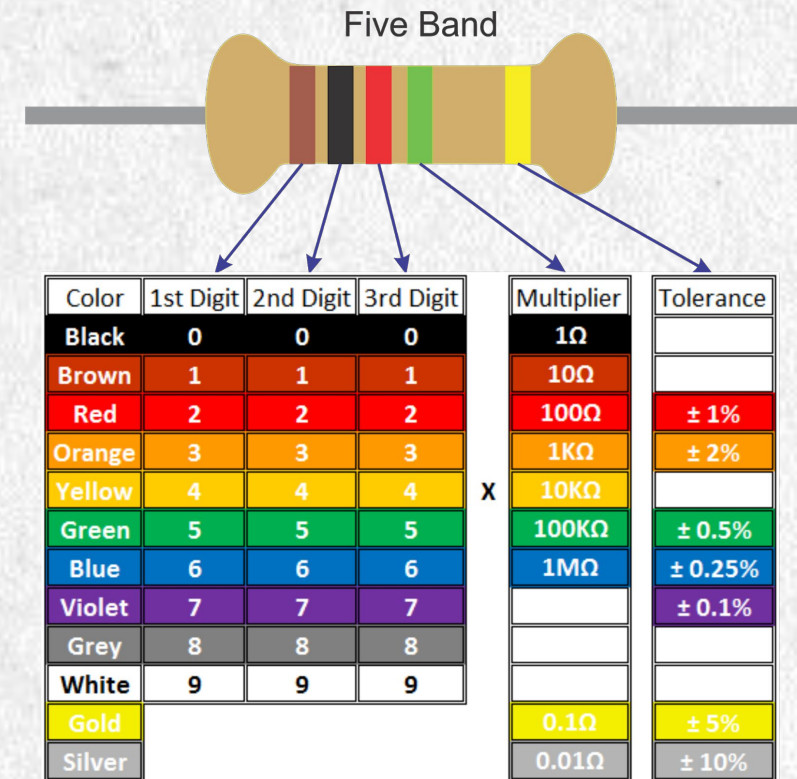
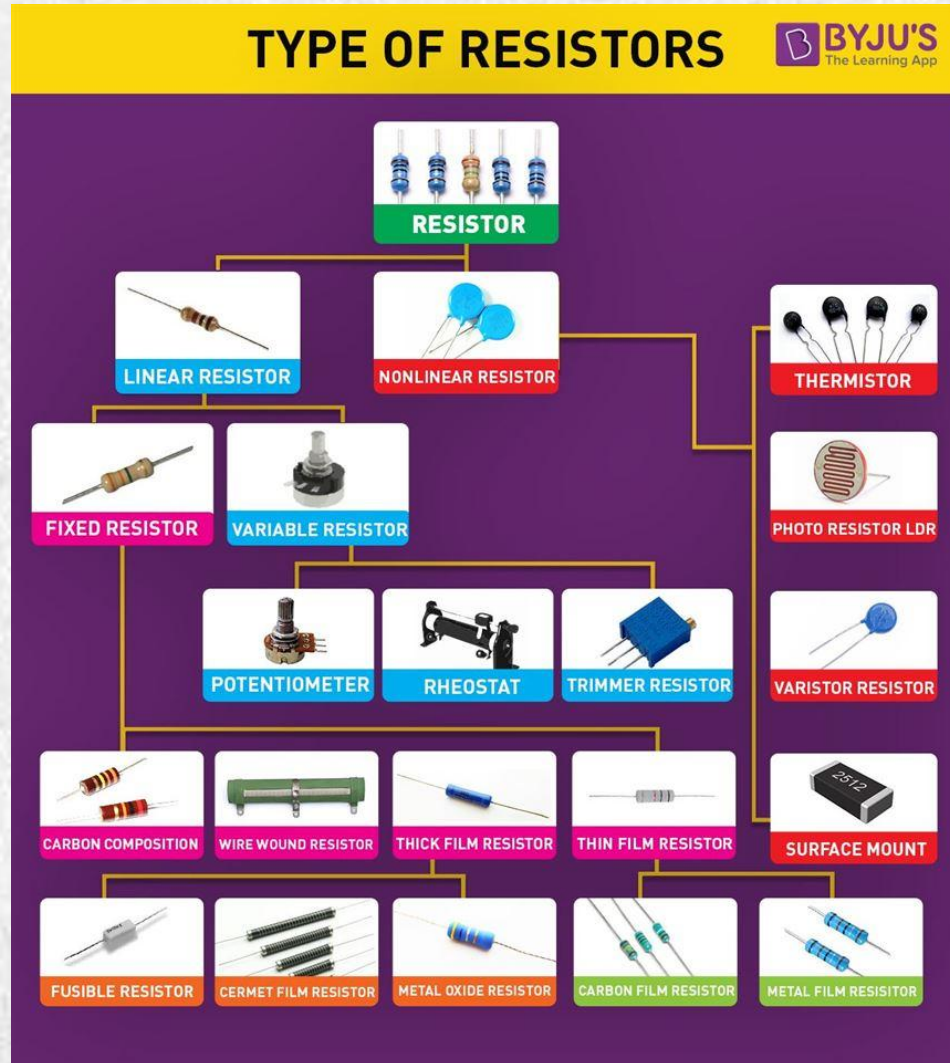
Female-to-Female



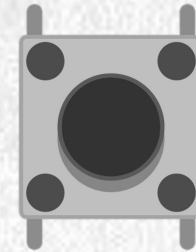
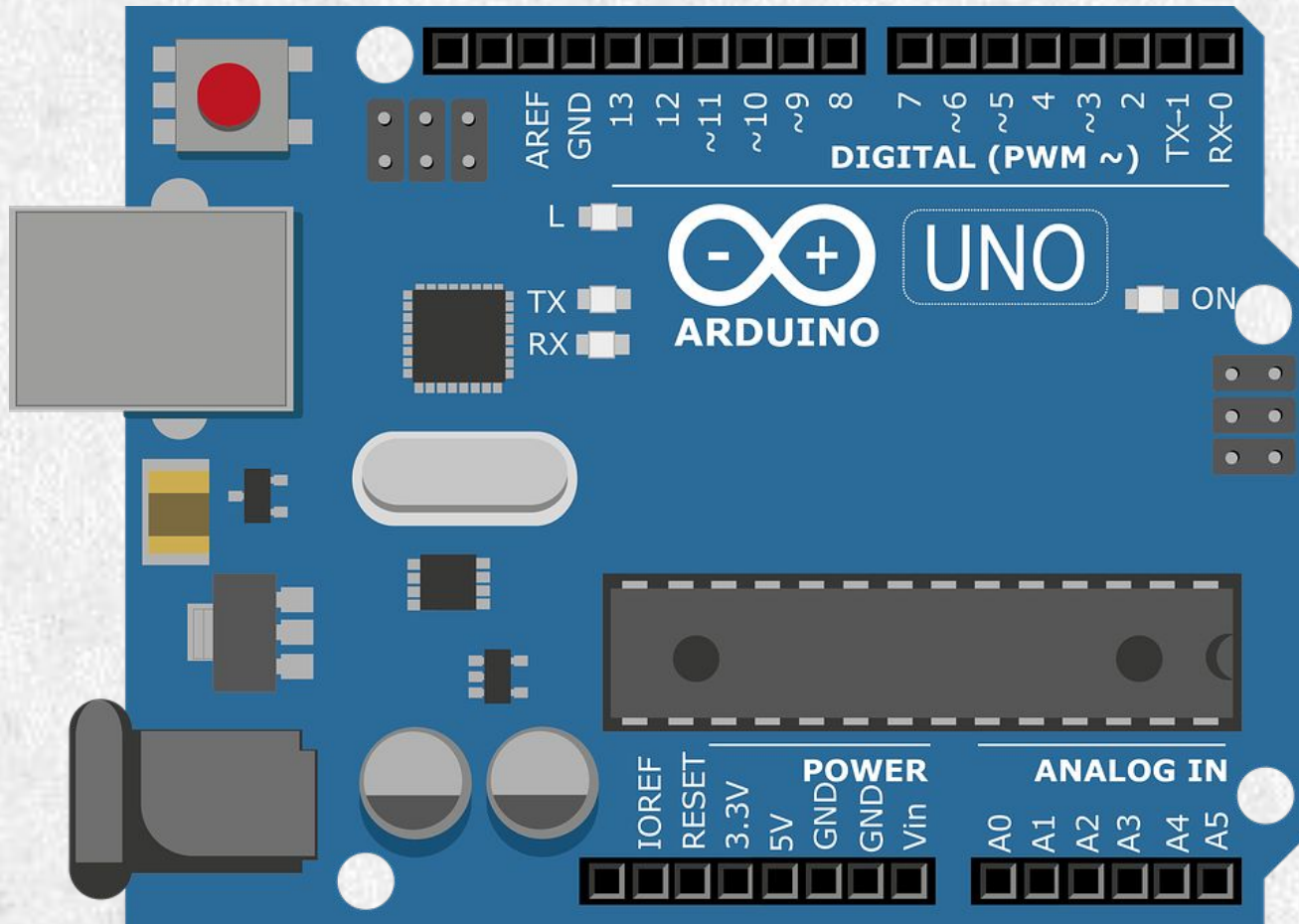
ANODE & CATHODE



RESISTORS



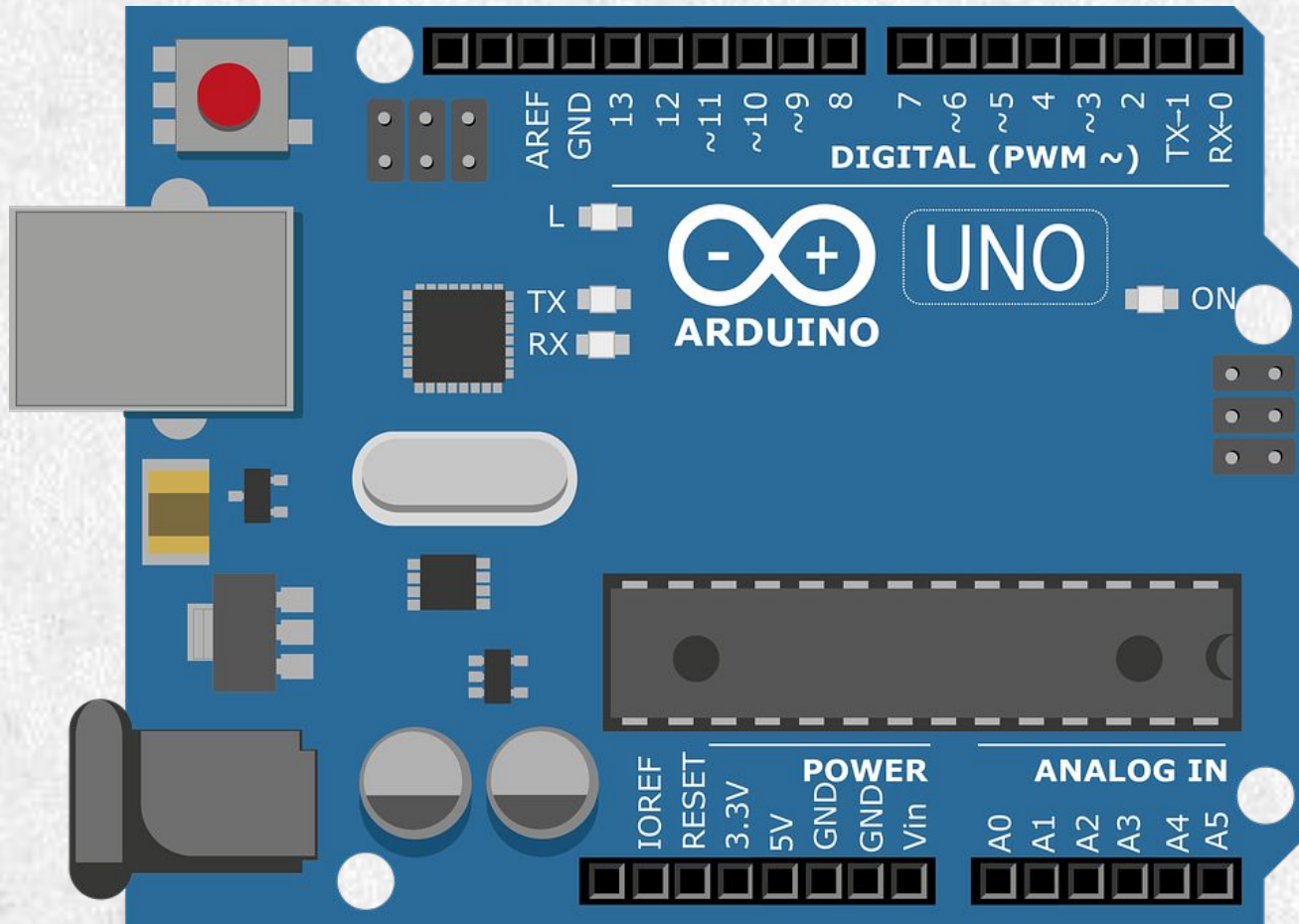
DIGITAL PINS



Digital Signal

```
1 = HIGH = true  
0 = LOW = false
```

DIGITAL PINS



```
#define var_name 5  
int var_name = 5;
```



INPUT

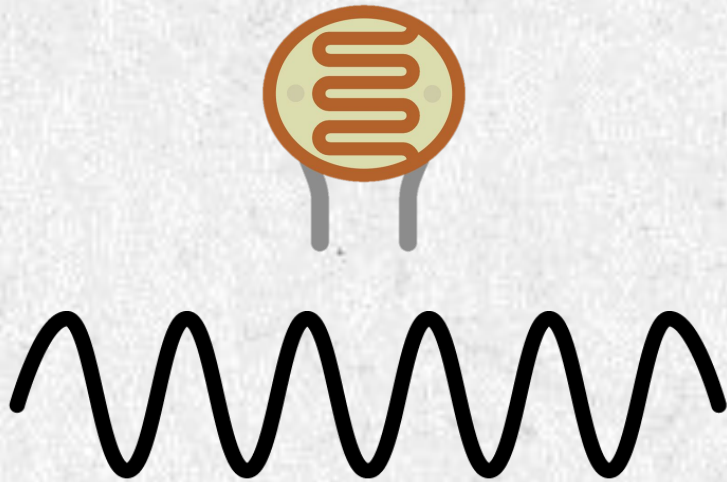
```
pinMode(var_name, INPUT);  
digitalRead(var_name);
```



OUTPUT

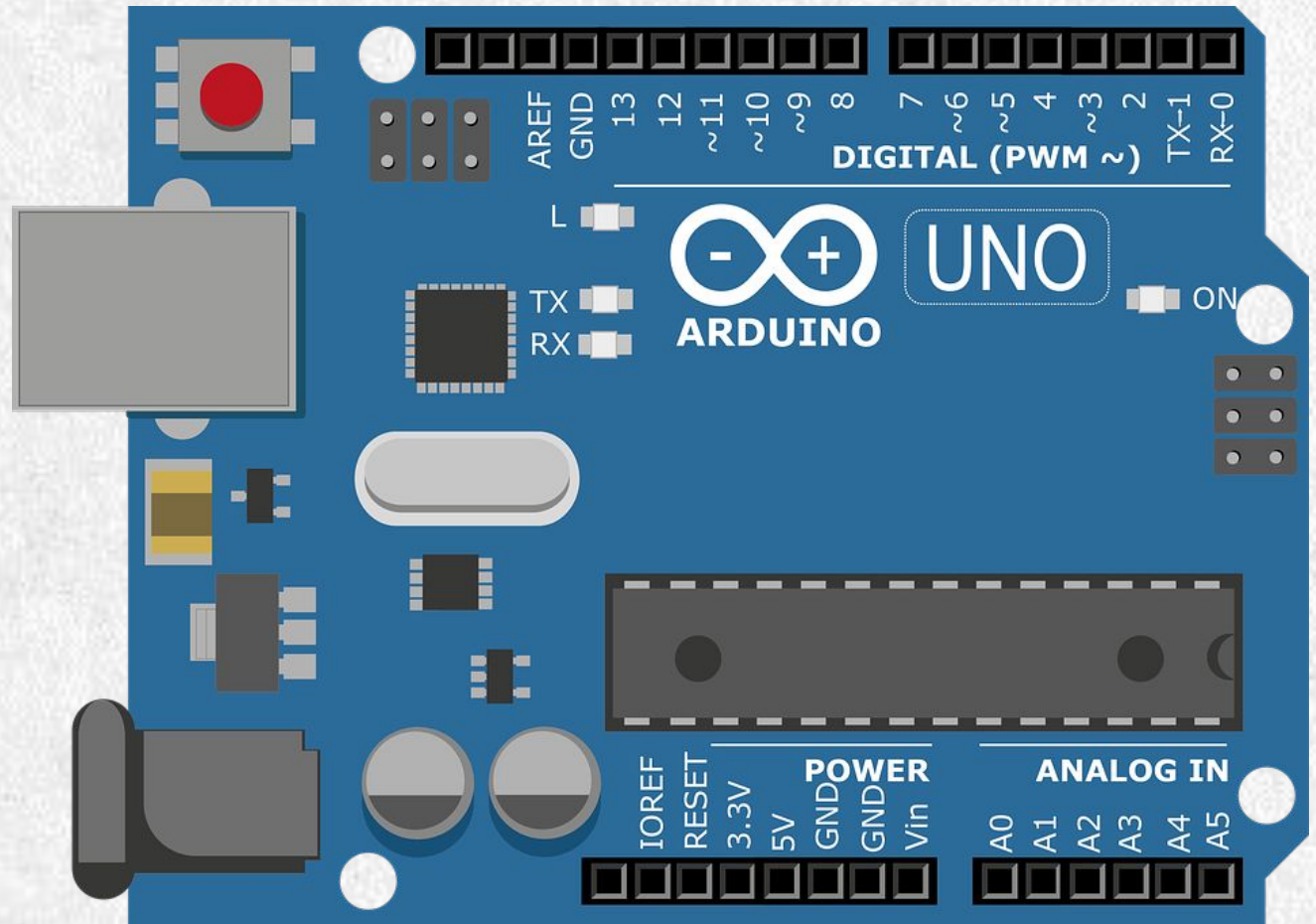
```
pinMode(var_name, OUTPUT);  
digitalWrite(#var_name, HIGH);  
digitalWrite(#var_name, LOW);
```

ANALOG PINS

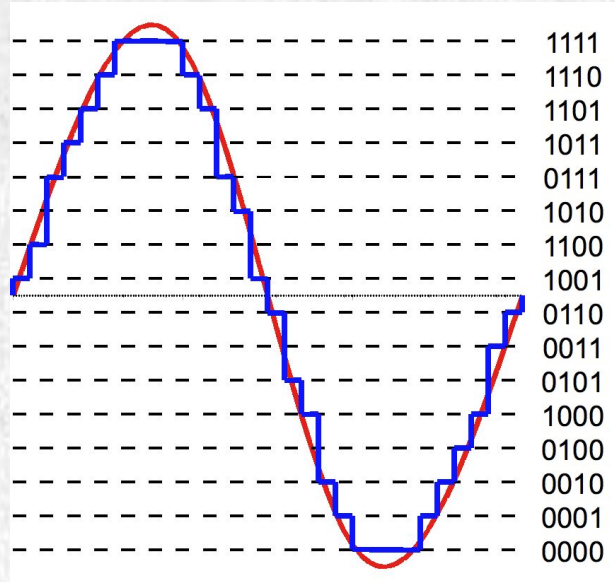


Analog Signal

```
#define var_name A0  
int var_name = A0;  
  
analogRead(var_name);
```



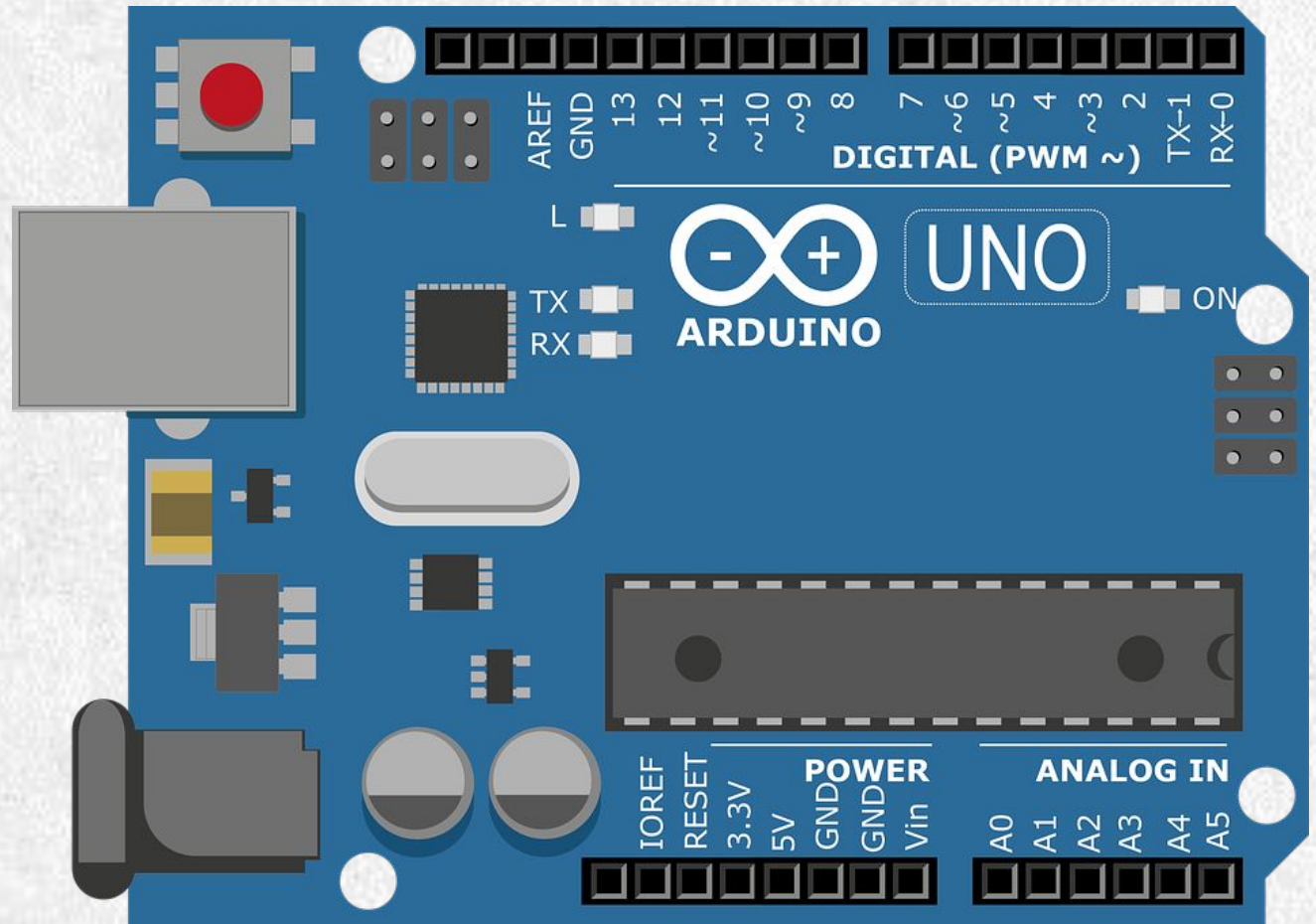
ANALOG PINS



The Arduino has a built-in 10-bit Analog-to-Digital Converter (ADC).

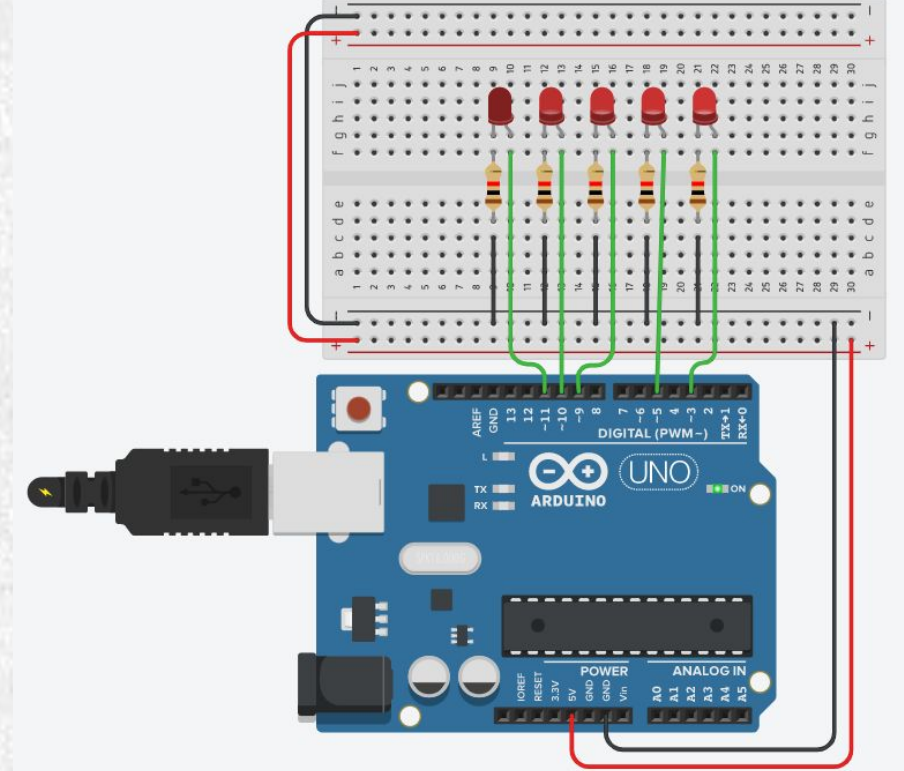
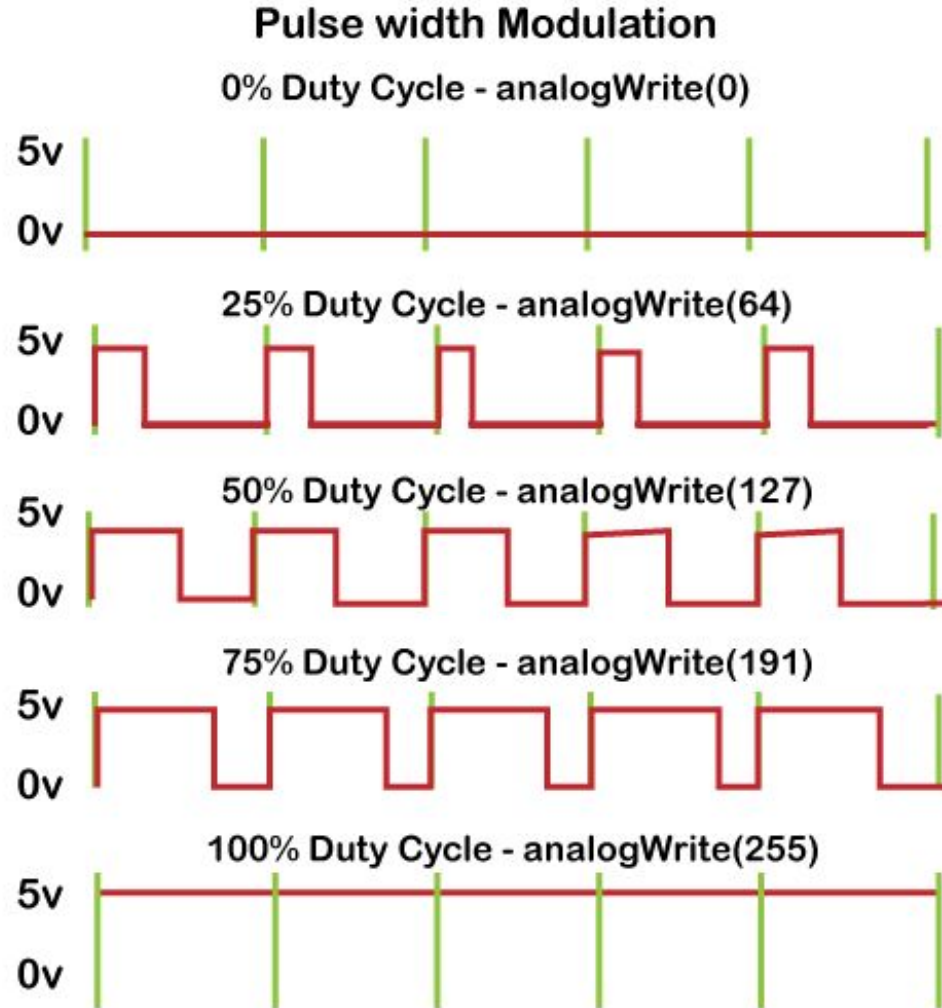
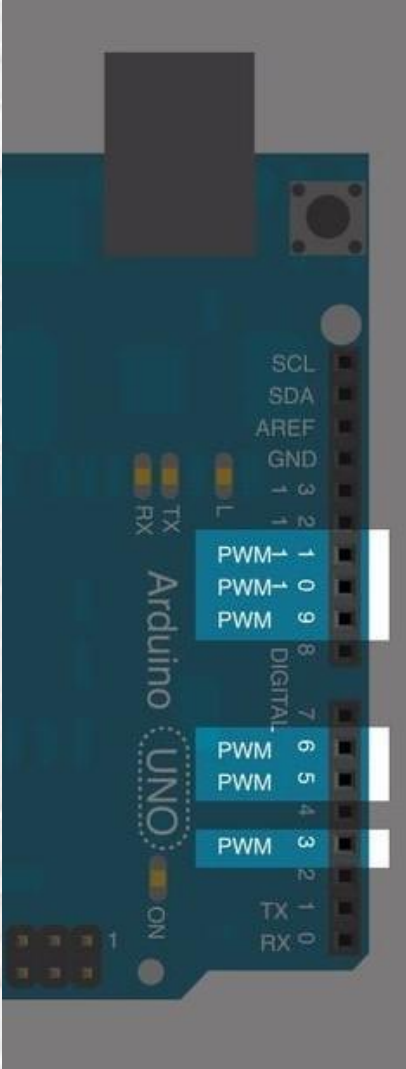
$$2^{10} = 1024$$

This means that it will map input voltages between 0 and the operating voltage (5V or 3.3V) into integer values between 0 and 1023.



WHAT ABOUT GETTING ANALOG OUTPUT?

2



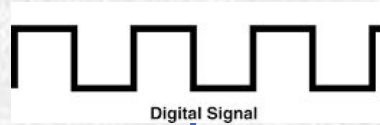
```
#define var_name 10
int var_name = 10;
pinMode(var_name, OUTPUT);
analogWrite(var_name, 64);
```

INPUT



- pinMode(pin, INPUT);
- pinMode(pin, INPUT_PULLUP);
- **digitalRead**(pin)
- Returns 0 or 1

DIGITAL



OUTPUT

- pinMode(pin, OUTPUT);
- pinMode(pin, INPUT);
- **digitalWrite**(pin, 0 or 1)
- Write 0 or 1



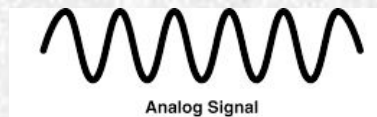
INPUT



- **analogRead**(pin)
- Returns 0 - 1023

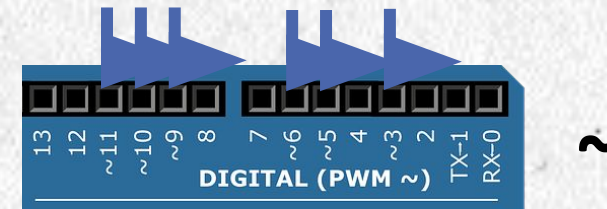


ANALOG

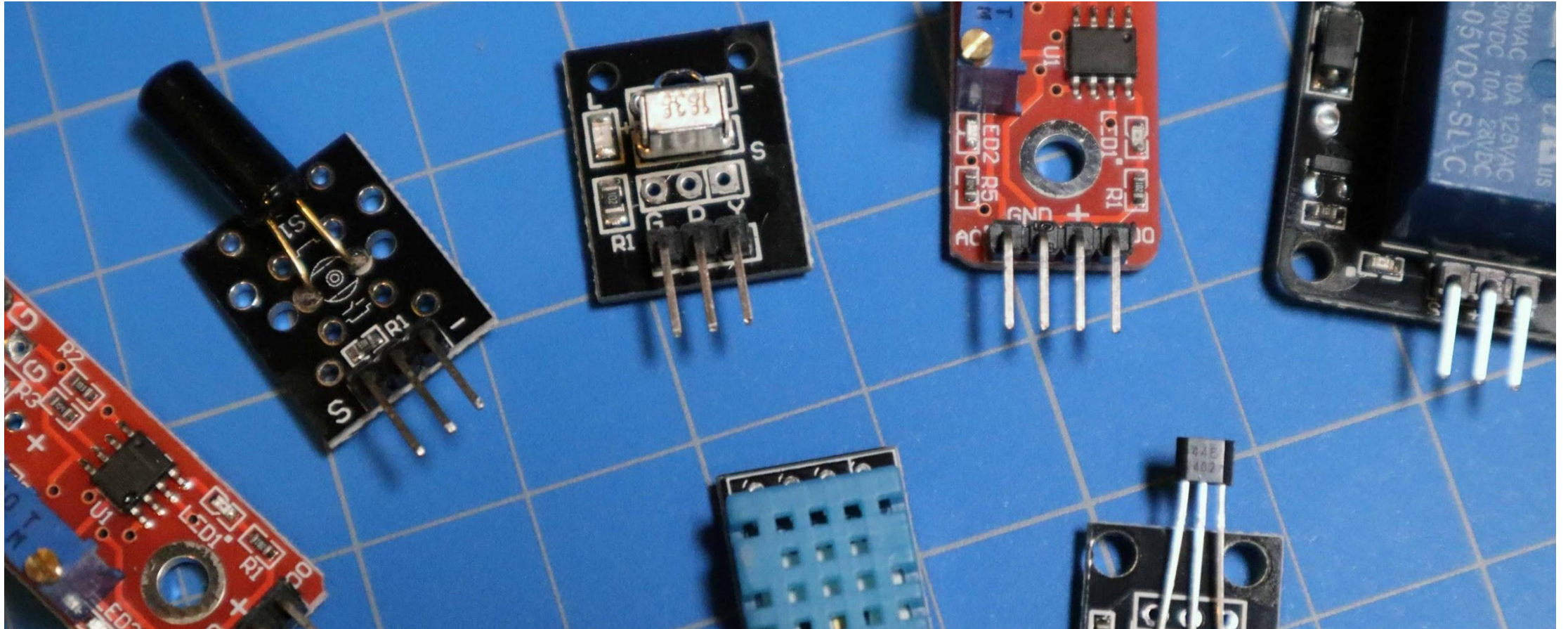


OUTPUT

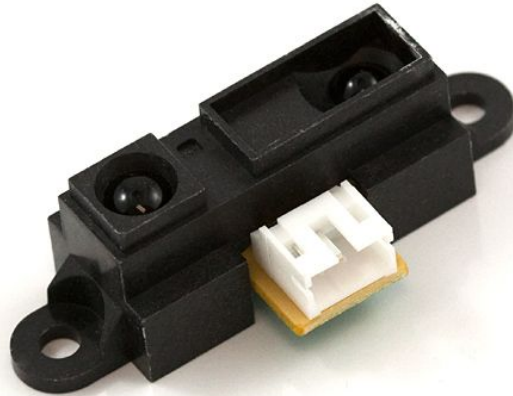
- **analogWrite**(pin, 0-255)
- Write 0 - 255



MANY KINDS OF SENSORS



Types of Obstacle Sensors



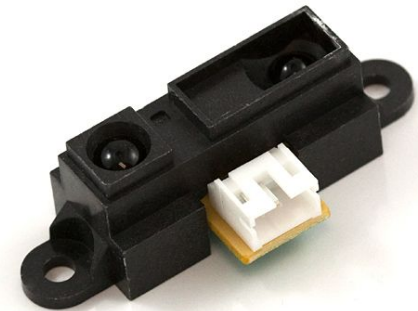
Types of Obstacle Sensors



3-80 cm
Digital Signal
High speed

Light Speed =
299,792,458 m/s

20-150 cm
Analog Signal
High speed



2cm-4m
Very slow
Sound Speed = 343 m/s

Obstacle Sensor (IR Sensor)



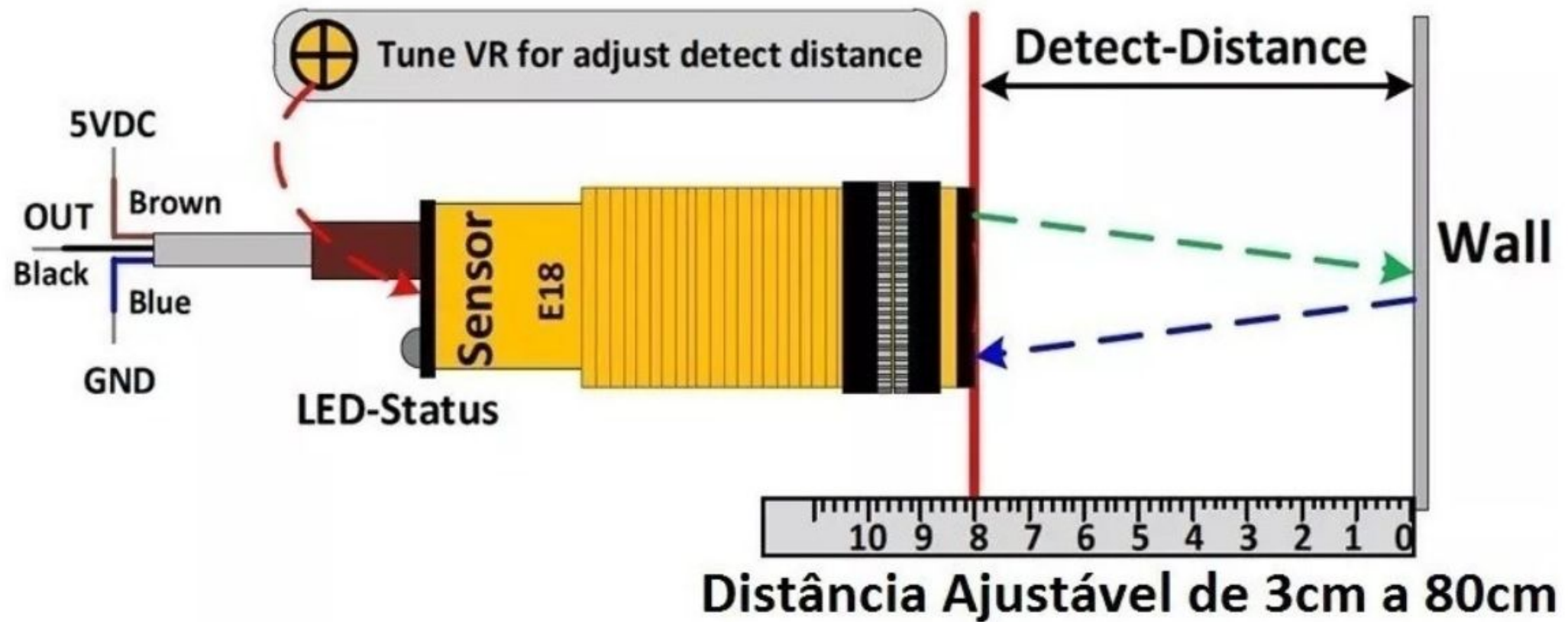
The E18-D80NK adjustable IR Infrared Proximity Sensor Switch features an IR sensor that can measure distance between 3 to 80 cm. The infrared proximity switch module is a reflection-type photoelectric sensor which integrates transmitting and receiving infrared beams function. Infrared proximity switches work by sending out beams of invisible infrared light.

Obstacle Sensor (IR Sensor)



- **Input voltage:** 5V DC
- **Current consumption:** >25mA (min) > 100mA (max)
- **Dimension:** 1.7cm (diameter) x 4.5cm (length)
- **Cable length:** 45cm
- **Detection of objects:** transparent or Opaque
- **Sensing range:** 3cm-80cm
- **NPN output:** (normally high)
- **Environment temperature:** -25 degC ~ 55 degC

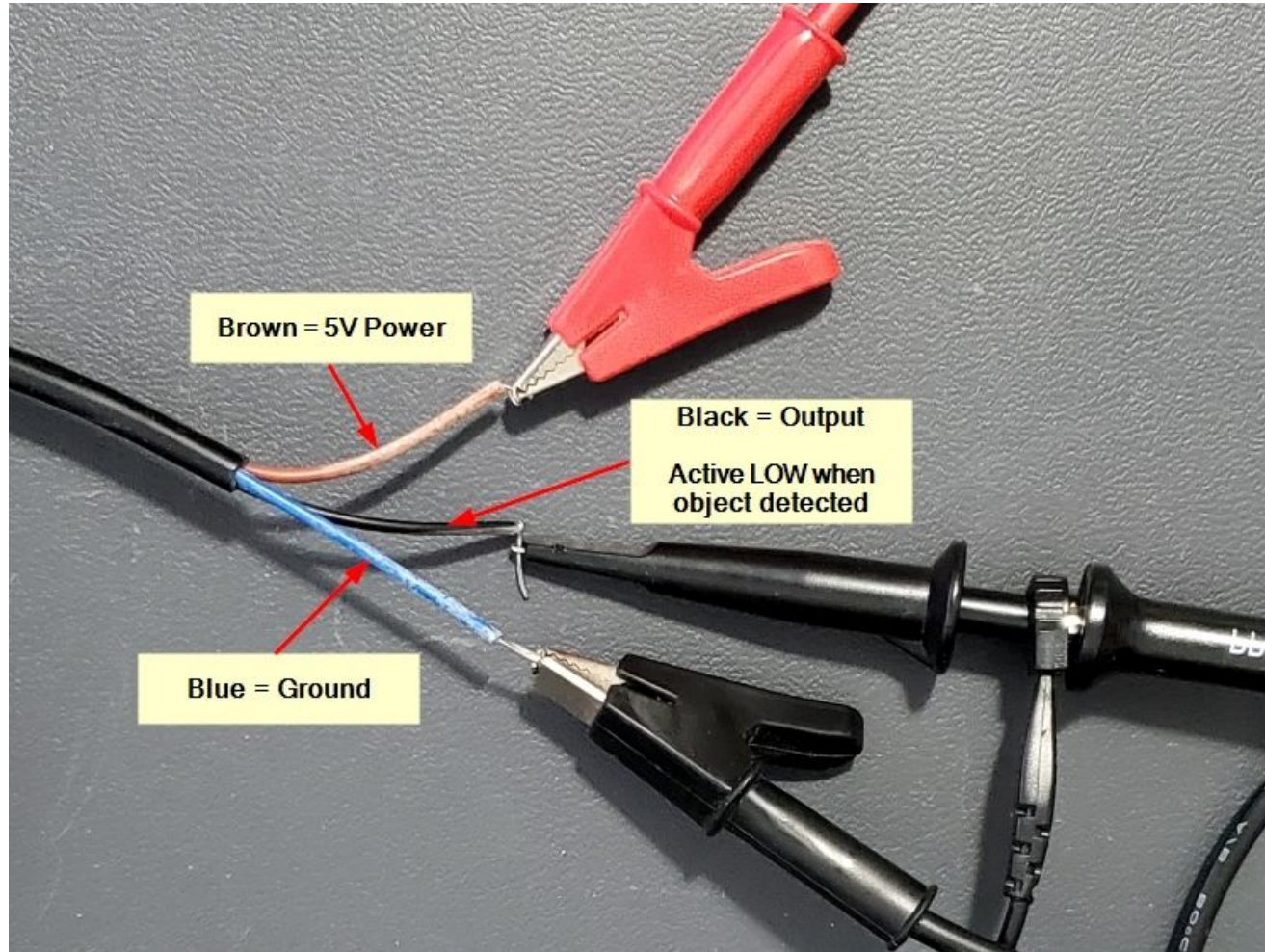
Obstacle Sensor (IR Sensor)



Obstacle Sensor (IR Sensor)



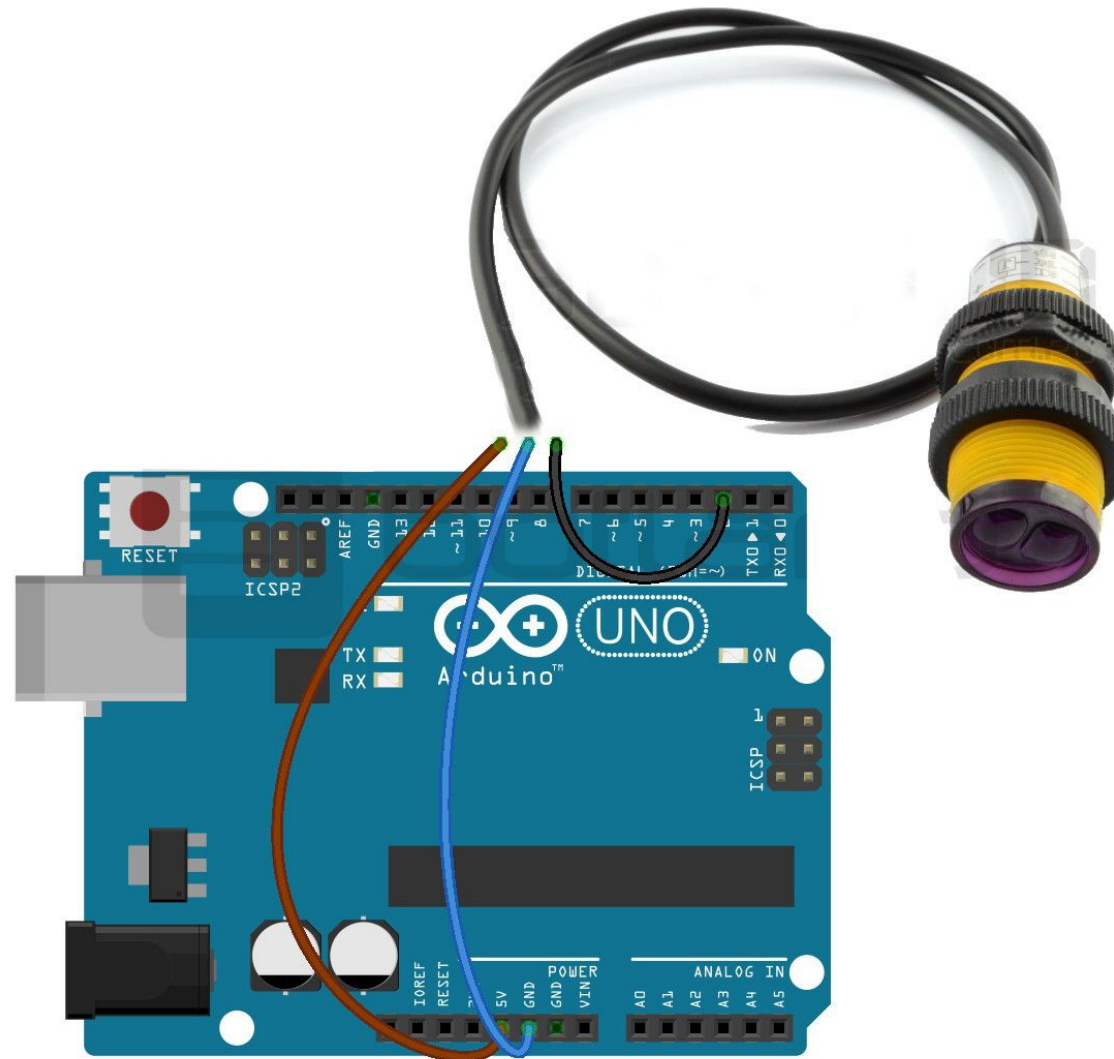
Obstacle Sensor (IR Sensor)



Obstacle Sensor (IR Sensor)



Obstacle Sensor (IR Sensor)



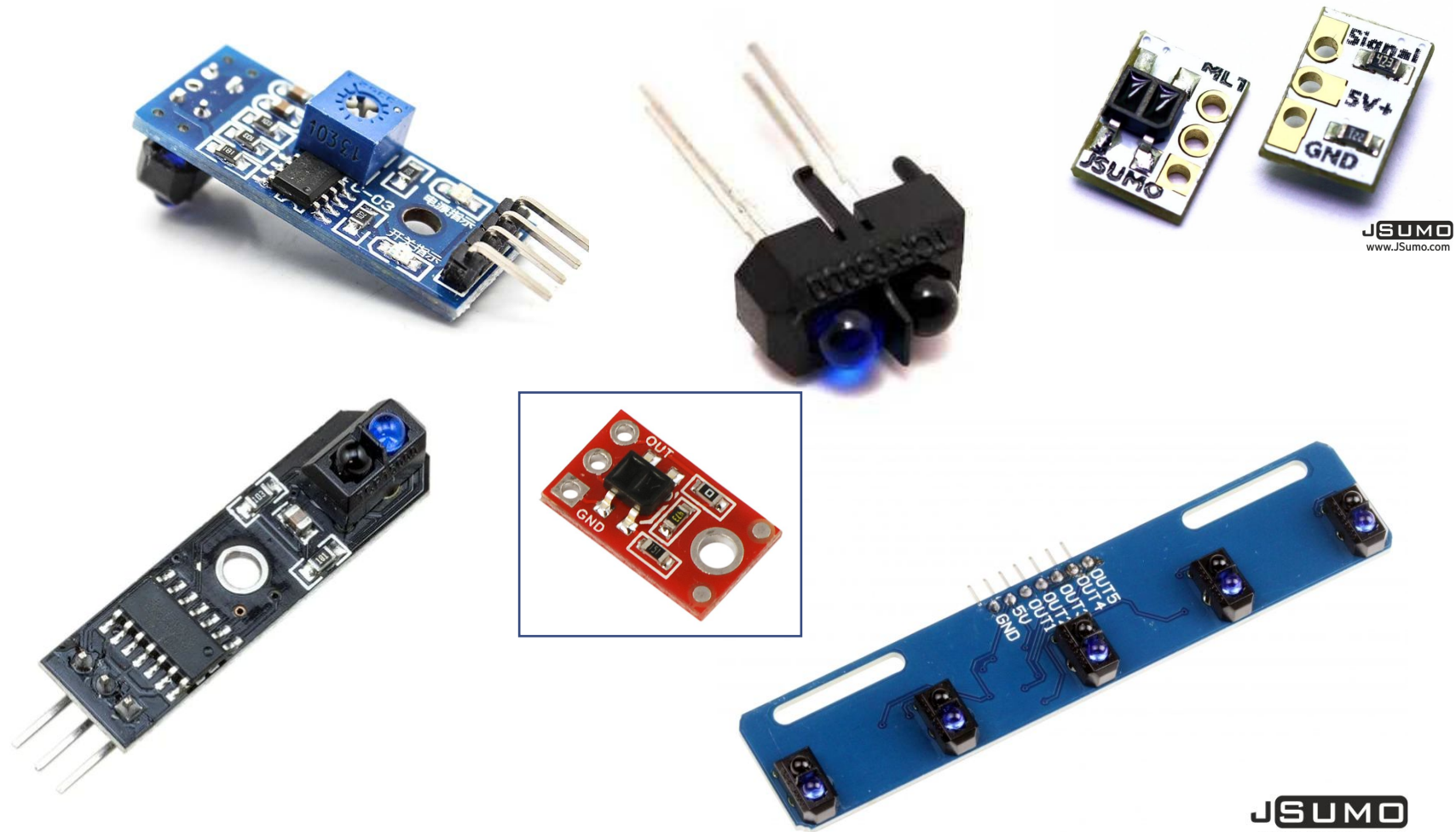
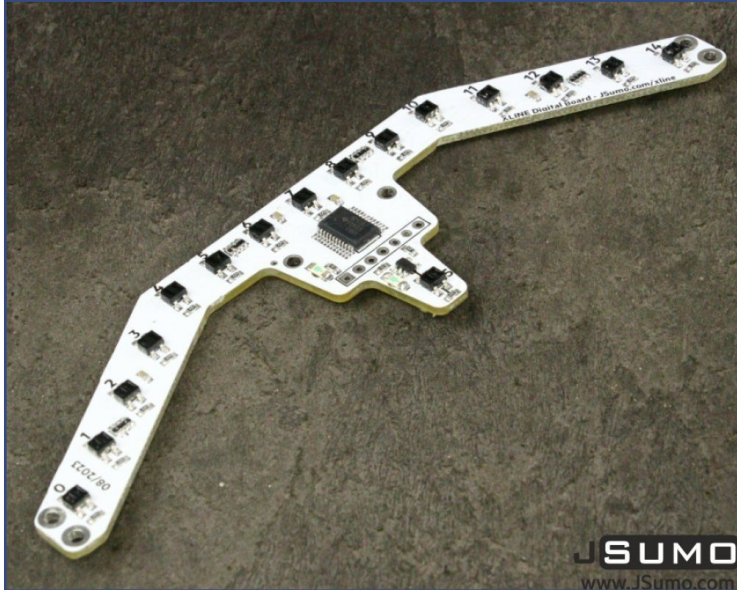
Obstacle Sensor (IR Sensor)

```
const int e18_sensor = 2;

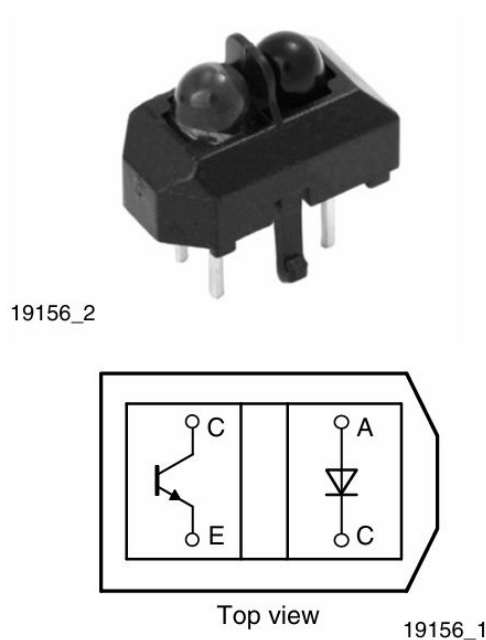
void setup() {
  Serial.begin(9600);
  pinMode (e18_sensor, INPUT);
}

void loop() {
  int value = digitalRead(e18_sensor);
  Serial.println(value);
  if(value==LOW)
  |   Serial.println("Object Detected");
  else
  |   Serial.println("All Clear");
}
```

Types of Black and White Sensors

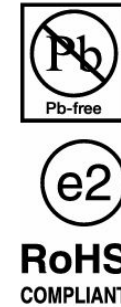


Types of Black and White Sensors



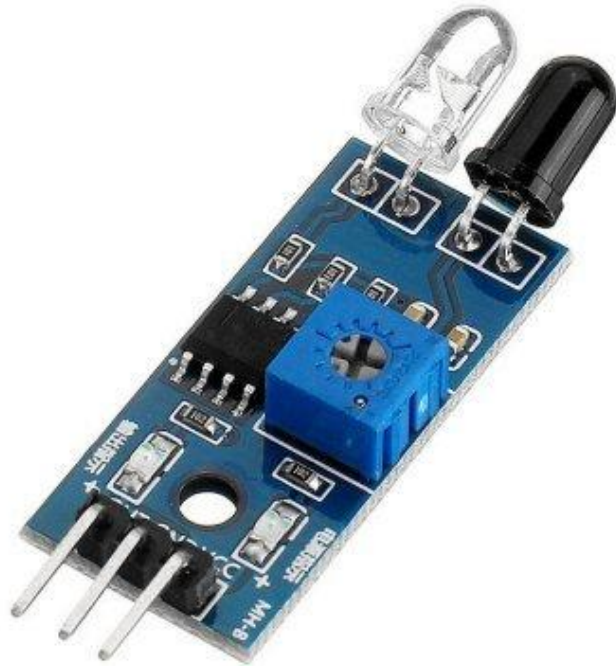
FEATURES

- Package type: leaded
- Detector type: phototransistor
- Dimensions (L x W x H in mm): 10.2 x 5.8 x 7
- Peak operating distance: 2.5 mm
- Operating range within > 20 % relative collector current: 0.2 mm to 15 mm
- Typical output current under test: $I_C = 1 \text{ mA}$
- Daylight blocking filter
- Emitter wavelength: 950 nm
- Lead (Pb)-free soldering released
- Compliant to RoHS directive 2002/95/EC and in accordance to WEEE 2002/96/EC



Response Time - Distance Range - Weight - Dimensions

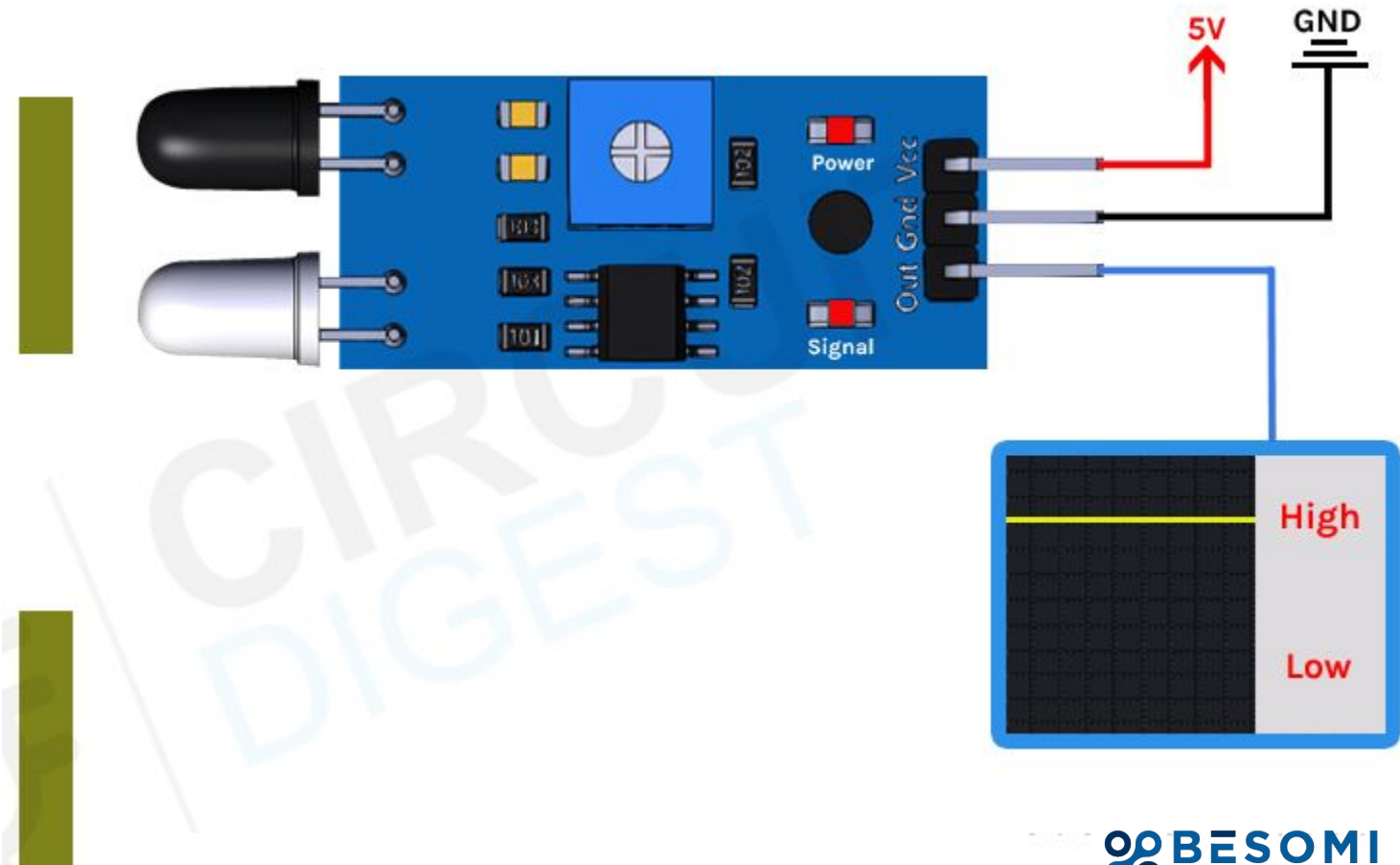
Black and White Sensor (IR SENSOR)



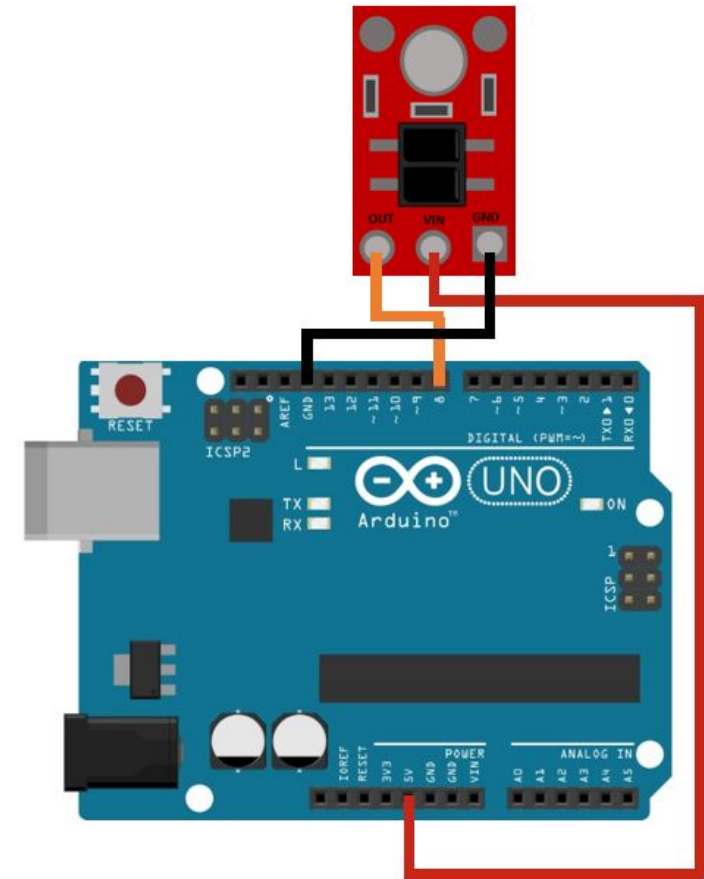
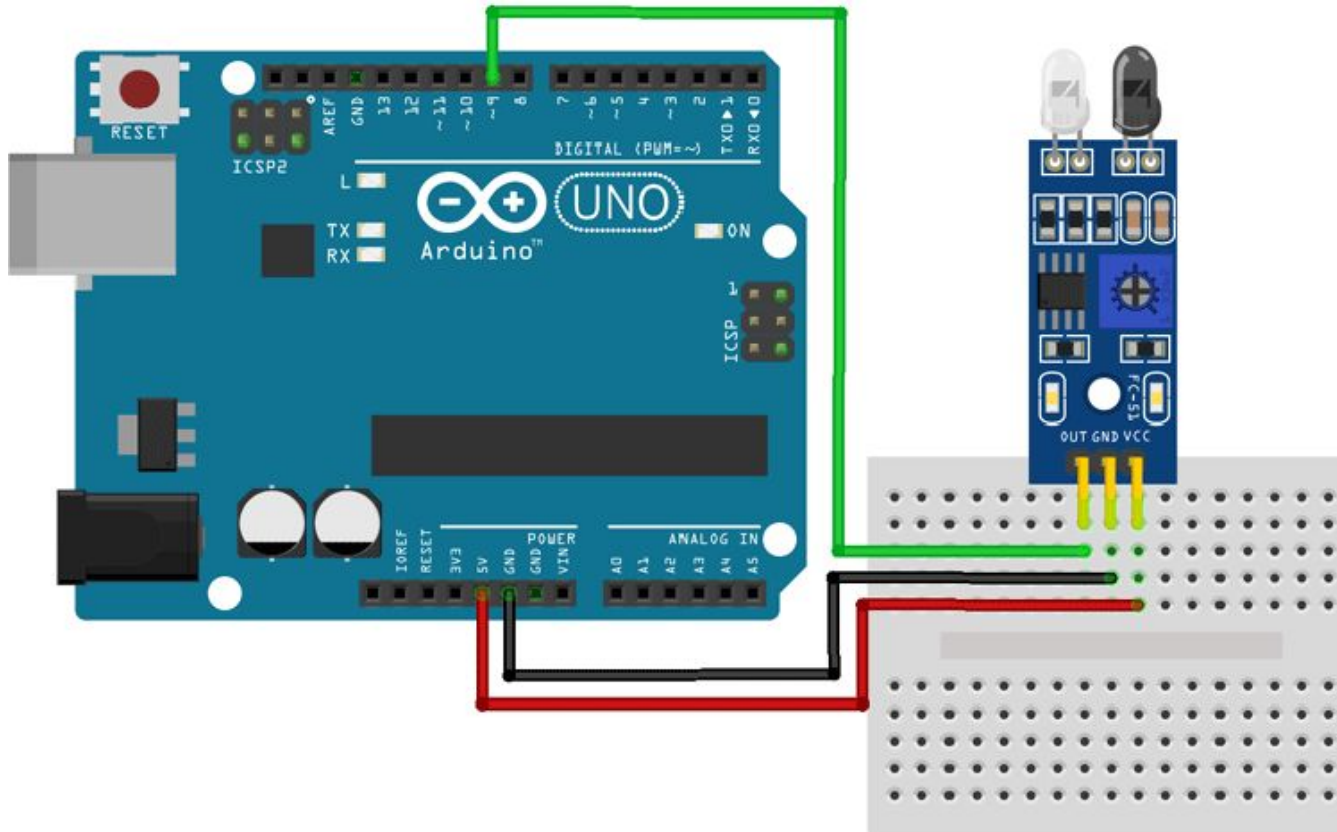
An **infrared proximity sensor** or **IR Sensor** is an electronic device that emits infrared lights to sense some aspect of the surroundings and can be employed to detect the motion of an object.

Black and White Sensor (IR SENSOR)

The OUT pin from the IR sensor outputs a **LOW** value when it receives a reflected IR signal, i.e. detected an object (**white**), and a **HIGH** value when it is not receiving a reflected IR signal, i.e. no object (**black**).



Black and White Sensor (IR SENSOR)



Black and White Sensor (IR SENSOR)

```
const int sensor = 9;

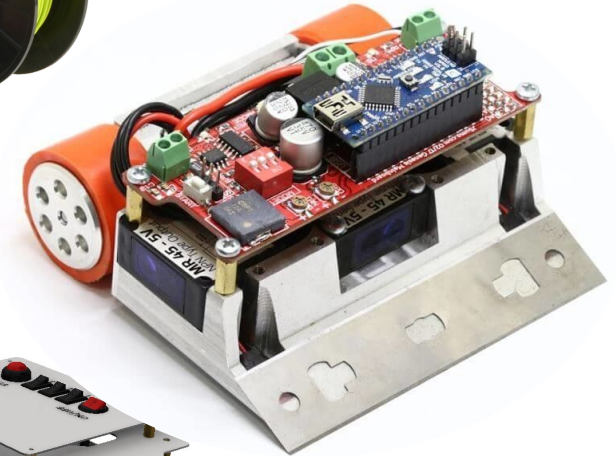
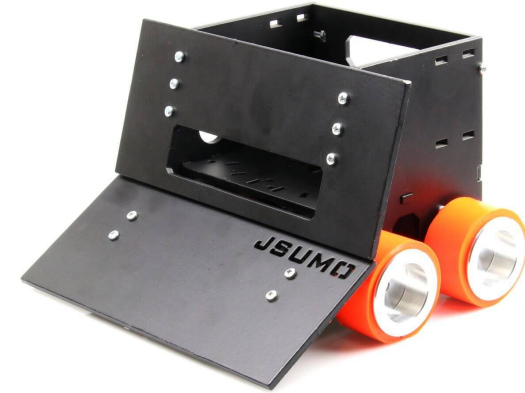
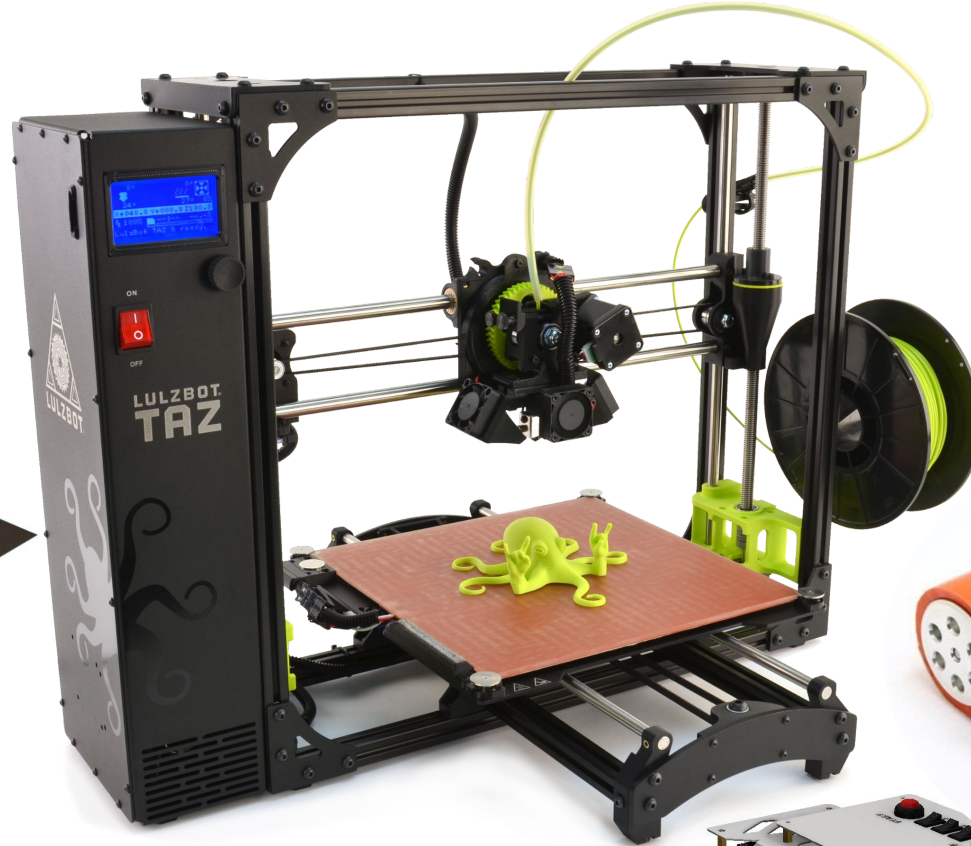
void setup() {
  Serial.begin(9600);
  pinMode (sensor, INPUT);
}

void loop() {
  int value = digitalRead(sensor);
  Serial.println(value);
  if(value==LOW)
  |   Serial.println("WHITE");
  else
  |   Serial.println("BLACK");
}
```

3D Design

3D Design

 AUTODESK[®]
FUSION 360[™]
 **SOLIDWORKS**



Fusion 360

Getting Started

To download and use Fusion 360, you will need an Autodesk ID. As a student or educator, you can obtain an Autodesk ID and Fusion 360 at

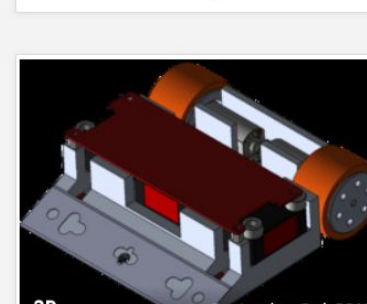
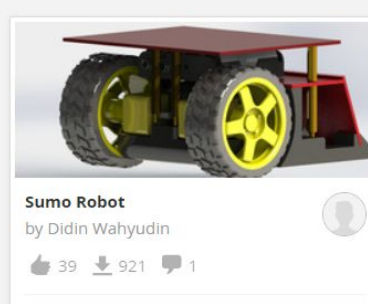
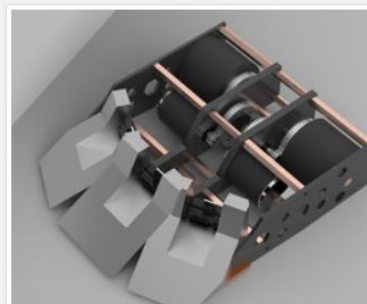
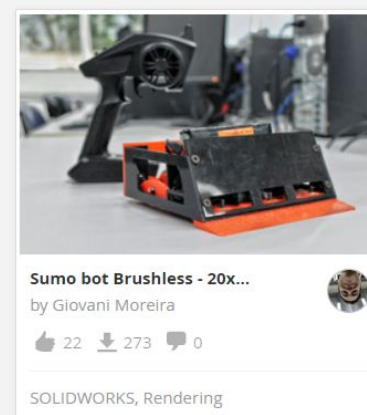
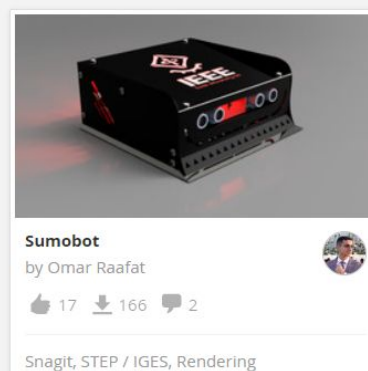
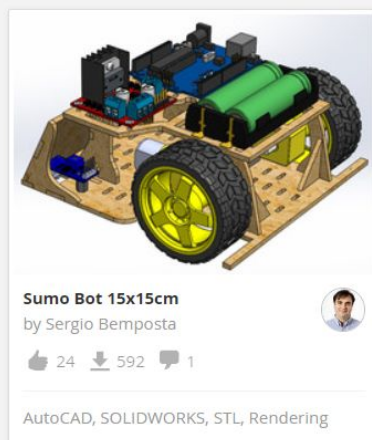
www.autodesk.com/education/free-software/fusion-360



Free to Use Designs

There are a lot of websites that let users download 3D designs for different components, or even a whole robot design. GRAB CAD are one of the most popular, with a big community.

GRABCAD



WHEELS

Type of wheels



Wheel Diameter



Small diameter:
MORE TORQUE



Big diameter:
MORE RPM

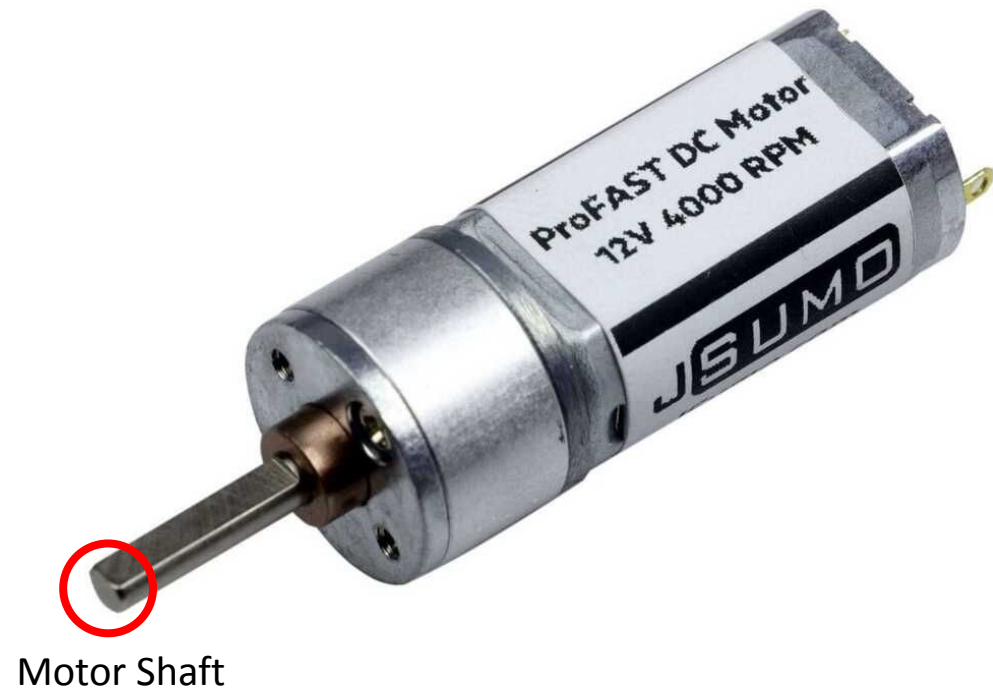
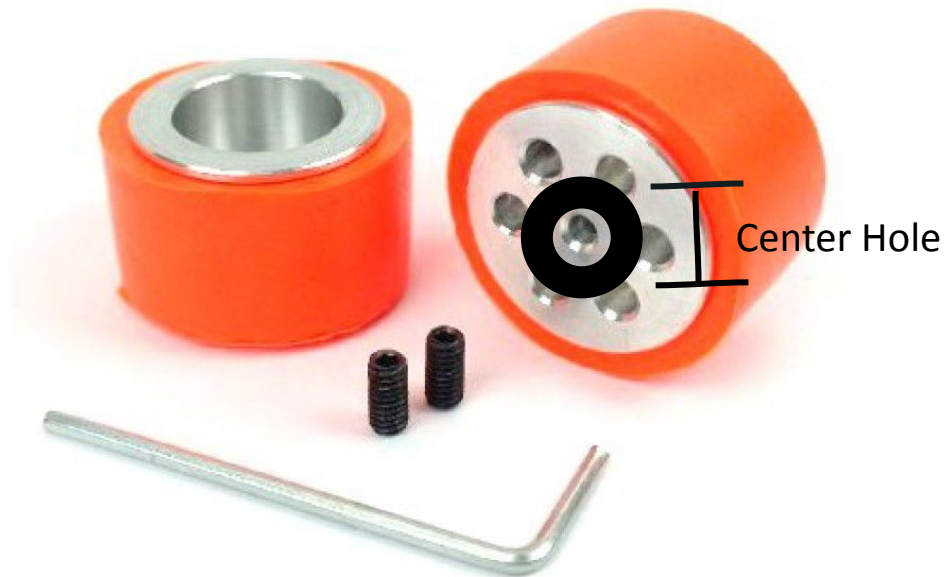
Choosing the best wheels

Which wheels are the best for my robot ?

Wheel Diameter

									
Diameter	26.5mm	26.5mm	33mm	33mm	38mm	37mm	43mm	52mm	74mm
Full Width	22mm	22mm	21mm	21mm	11mm	34mm	20mm	30mm	44mm
Silicone Width	19mm	19mm	18.5mm	18.5mm	8mm	30mm	17mm	28mm	41mm
Unit Weight (NOT PAIR)	13gr	25gr	22gr	52gr	18gr	51gr	26gr	48gr	229gr
Center Hole	3mm	3mm	3mm	3mm	3mm	3mm	4mm	6mm	6mm
Suitable for	Minisumo, Line Follower, Maze Robot	Minisumo, Maze Robot	Minisumo, Line Follower, Mazer Robot	Minisumo	Line Follower, General Projects	Line Follower, General Projects	General Robots, Non-Magnetic Sumo Robots	Non-Magnetic Sumo Robot, Explorer	Explorer Robots Bigger Projects

Center Hole / Shaft size



Easy Plug



BATTERIES

Battery Types

DIFFERENT TYPES OF BATTERIES



LITHIUM
POLYMER



COIN CELL



LEAD ACID



LITHIUM ION



NICKEL
CADMIUM



LITHIUM IRON
PHOSPHATE



BUTTON CELL



ALKALINE
BATTERIES

What we use in our robots ?



Lithium-Ion Batteries



Lithium Polymer

POWER THE MOTORS

Important things to look at when choosing your battery:

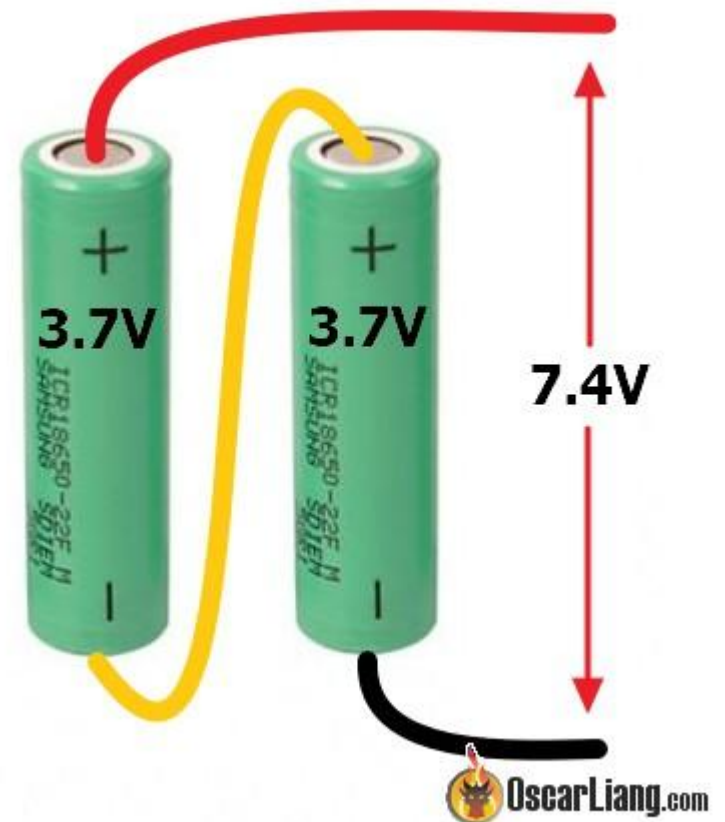
Voltage + ***Capacity***

POWER THE MOTORS



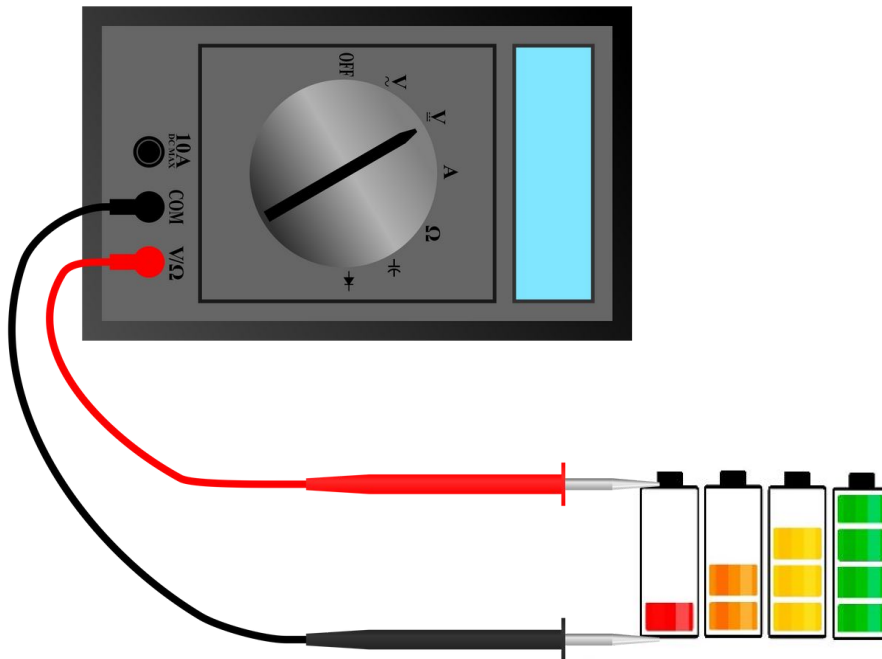
POWER THE MOTORS

Voltage	State of Charge
12.60+	100%
12.50	90%
12.42	80%
12.32	70%
12.20	60%
12.06	50%
11.90	40%
11.75	30%
11.58	20%
11.31	10%
10.50	0%



POWER THE MOTORS

3.7V LI-ION BATTERY



MOTORS

Types of Electric Motors

WWW.ELECTRICALTECHNOLOGY.ORG



AC Motors



DC Motor



Special Motors



Induction Motor



Shunt Motor



Stepper Motor



Synchronous Motor



Series Motor



Brushless Motor



Commutator Motor



PMDC Motor



Servo Motor



Wound Rotor Motor



Compound Motor



Universal Motor



Squirrel Cage Motor



Separately Excited Motor



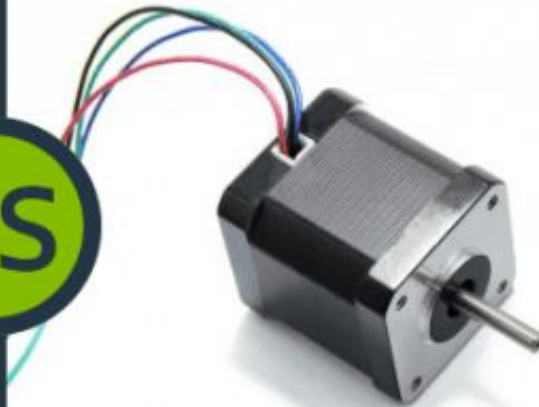
Reluctance Motor



The Difference Between



VS



VS



DC Motors

Stepper Motors

Servo Motors

Motors for SUMO Robot

How to choose the best motor ?

- Voltage Rating
- Stall Current
- RPM
- Torque



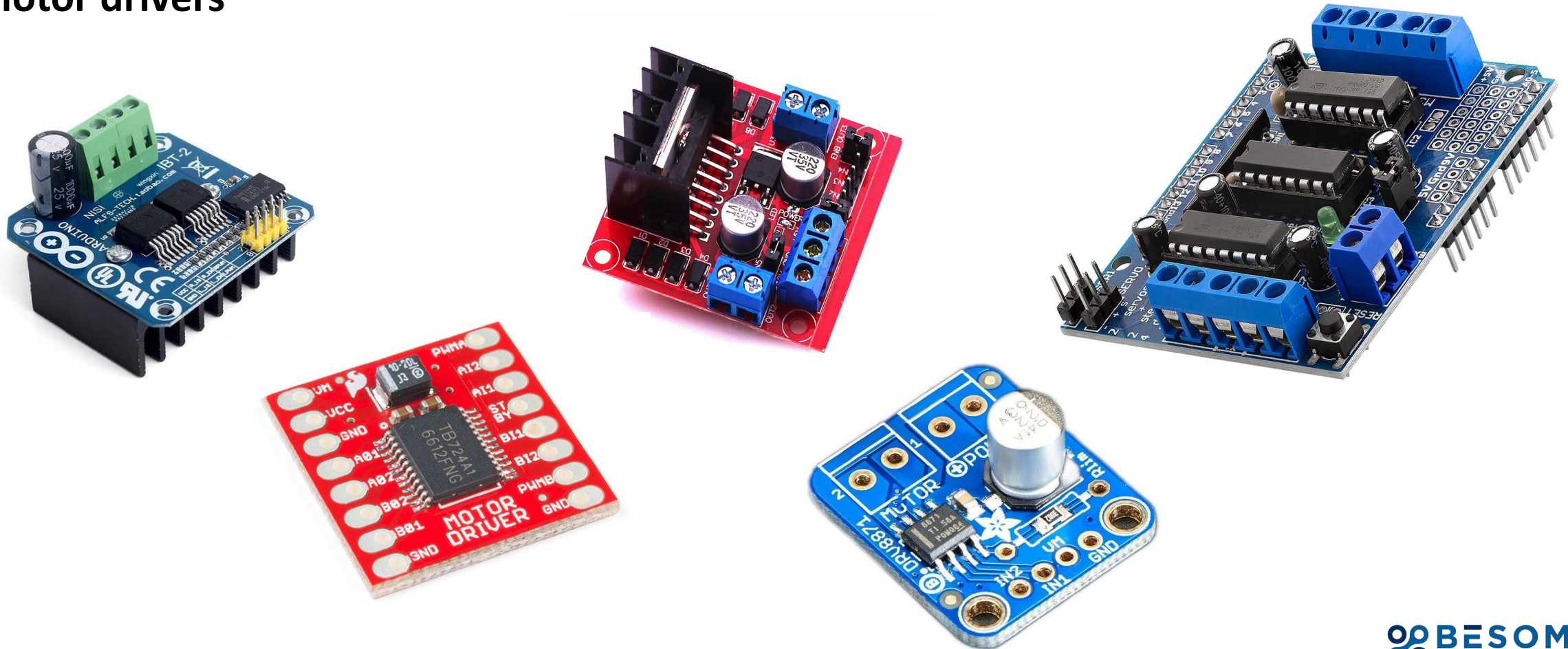
Motors for SUMO Robot

High Torque Motor or High Speed (RPM) Motor ?



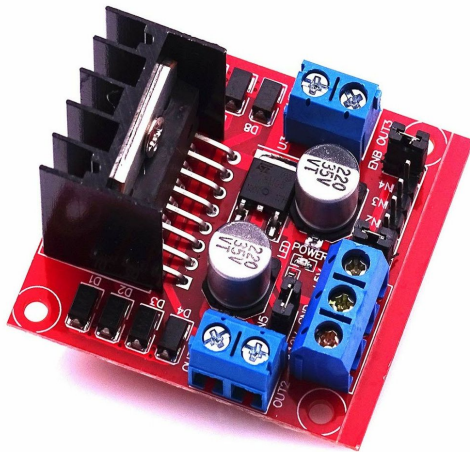
Motors for SUMO Robot

Motor drivers



Motors for SUMO Robot

Motor drivers



L298N Dual H-Bridge

Motor Voltage

5 - 35 Volts

Continuous Current

2 Amperes

Peak Current

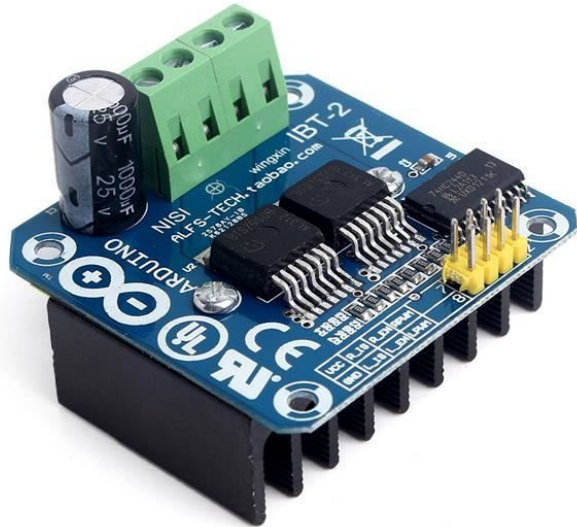
3.5 Amperes

Logic Voltage

3.3 - 5 Volts

Motors for SUMO Robot

Motor drivers



IBT2 (BTS7960) H-Bridge

Motor Voltage

6 - 27 Volts

Continuous Current

30 Amperes

Peak Current

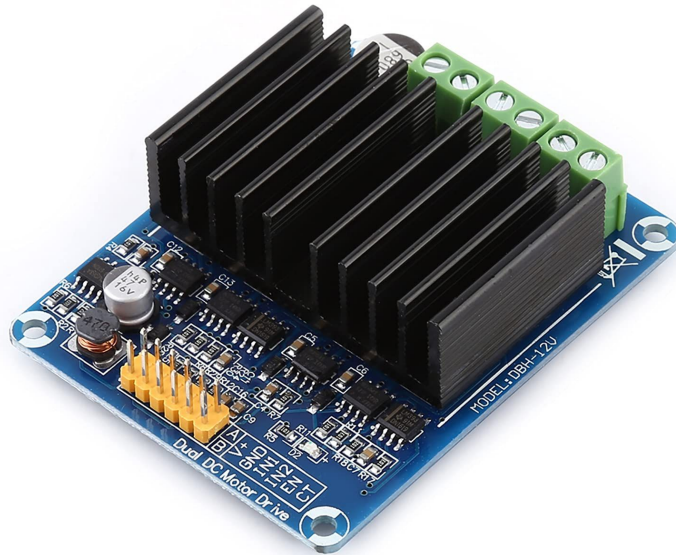
43 Amperes

Logic Voltage

3.3 - 5 Volts

Motors for SUMO Robot

Motor drivers



DBH-12 H-Bridge

Motor Voltage

5 - 14.5 Volts

Continuous Current

20 Amperes

Peak Current

30 Amperes

Logic Voltage

3.3 - 12 Volts

BTS7960 MOTOR DRIVER WITH DC MOTOR

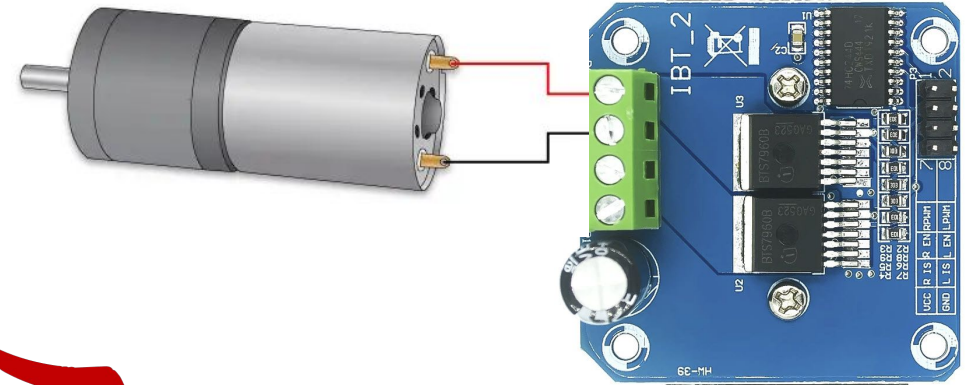
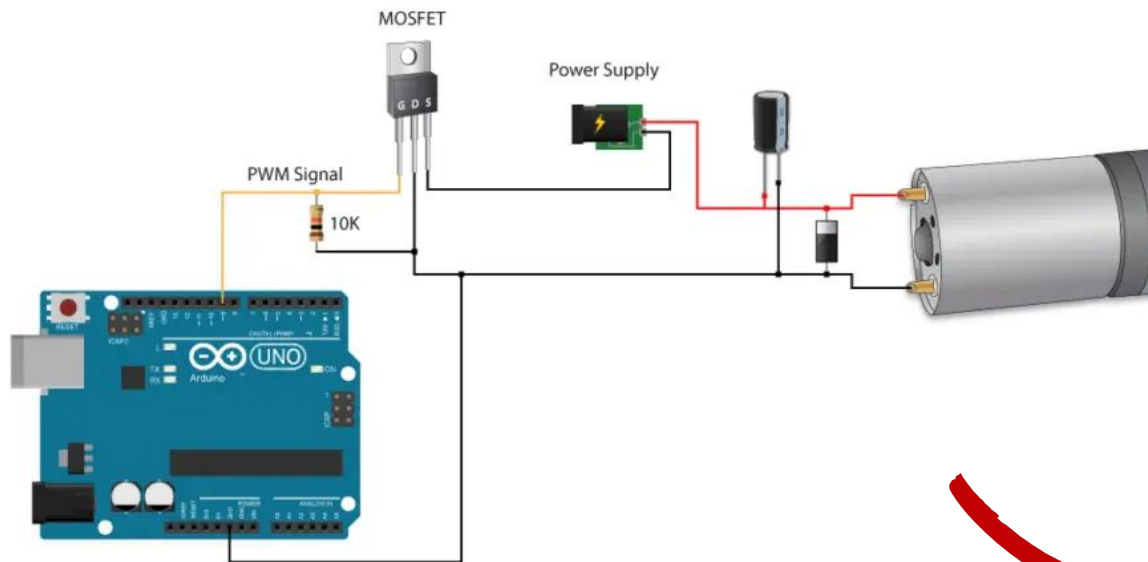
Controlling a DC Motor

In order to have a complete control over DC motor, we have to control its speed and rotation direction. This can be achieved by combining these two techniques.

- **PWM** – For controlling speed
- **H-Bridge** – For controlling rotation direction

BTS7960 MOTOR DRIVER WITH DC MOTOR

makes life easier!!

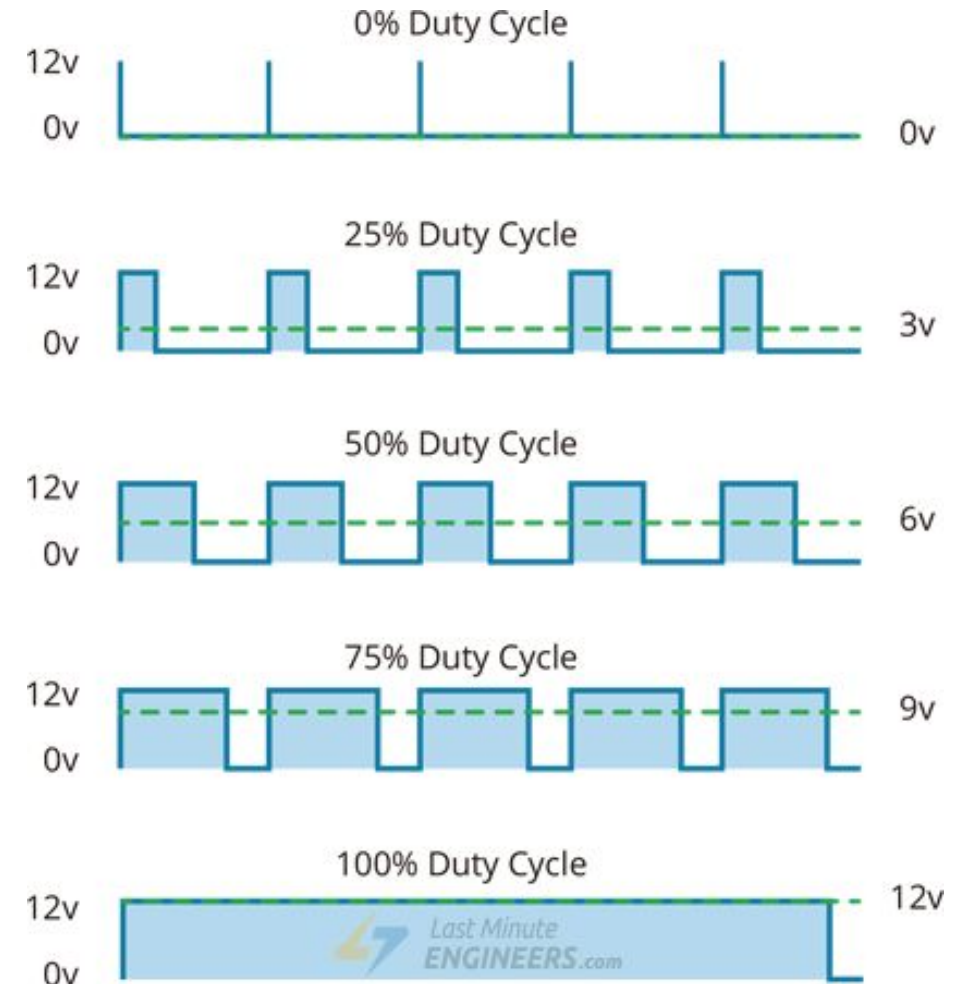


BTS7960 MOTOR DRIVER WITH DC MOTOR

A) PWM – For controlling speed

The speed of a DC motor can be controlled by varying its input voltage. A common technique for doing this is to use PWM (Pulse Width Modulation).

PWM is a technique where average value of the input voltage is adjusted by sending a series of ON-OFF pulses.

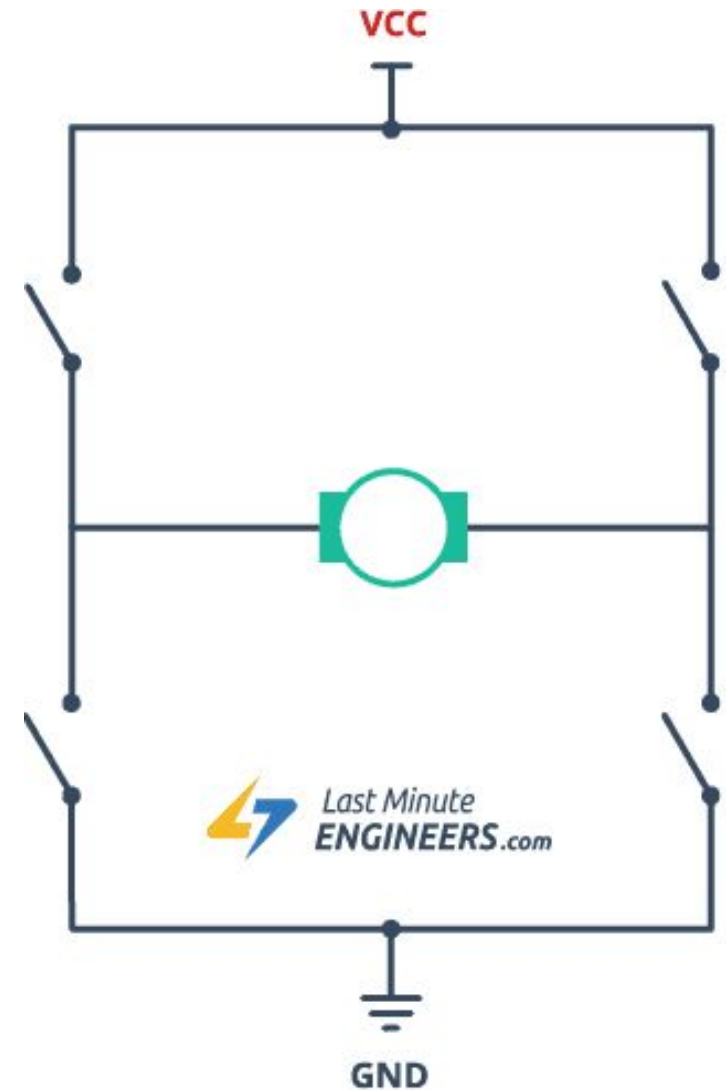


BTS7960 MOTOR DRIVER WITH DC MOTOR

B) H-Bridge – For controlling rotation direction

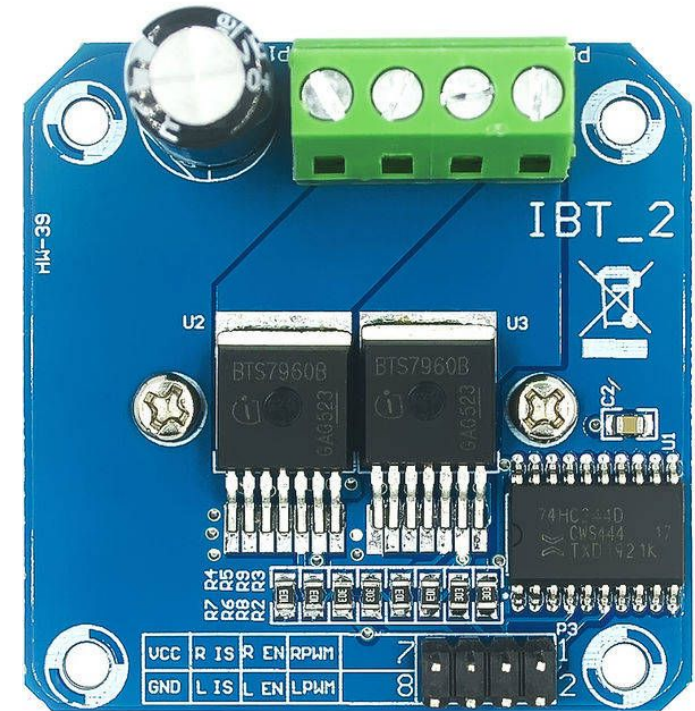
The DC motor's spinning direction can be controlled by changing polarity of its input voltage. A common technique for doing this is to use an H-Bridge.

An H-Bridge circuit contains four switches with the motor at the center forming an H-like arrangement.

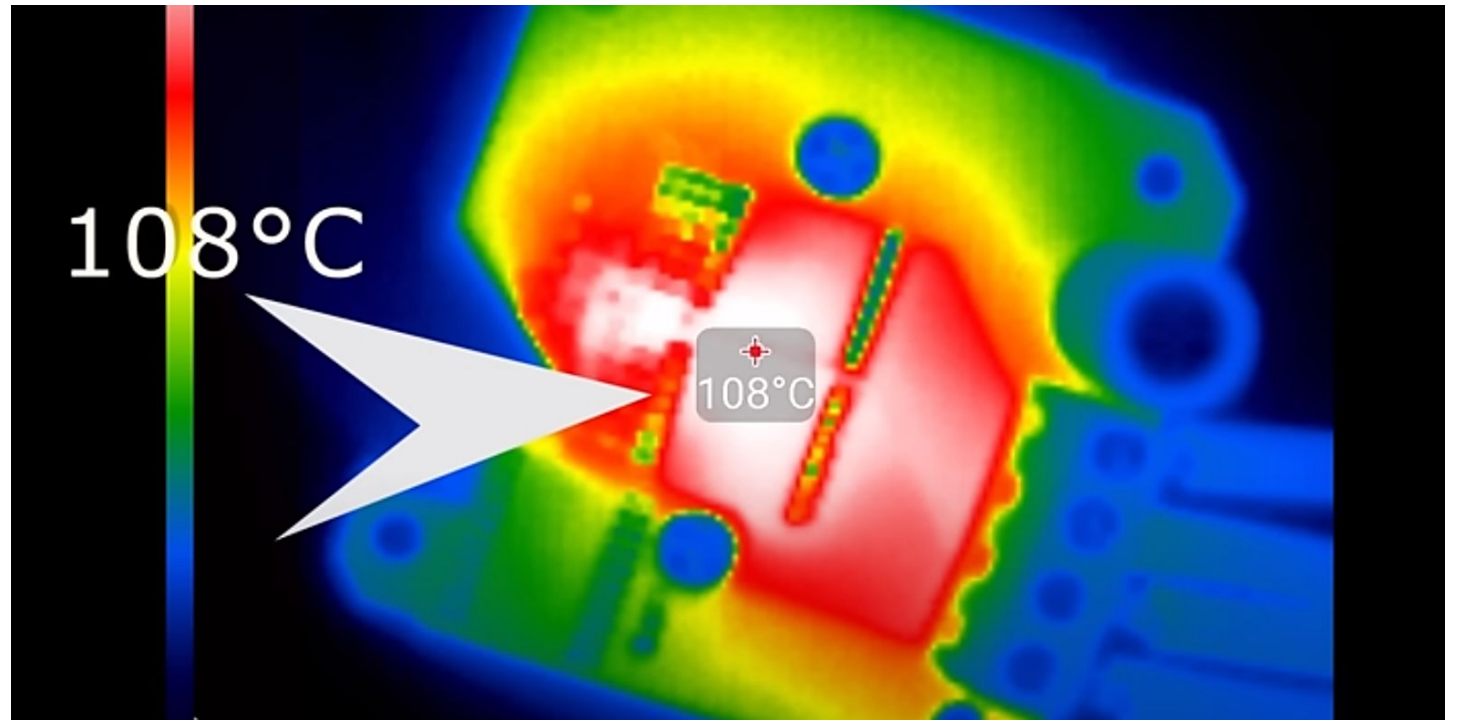
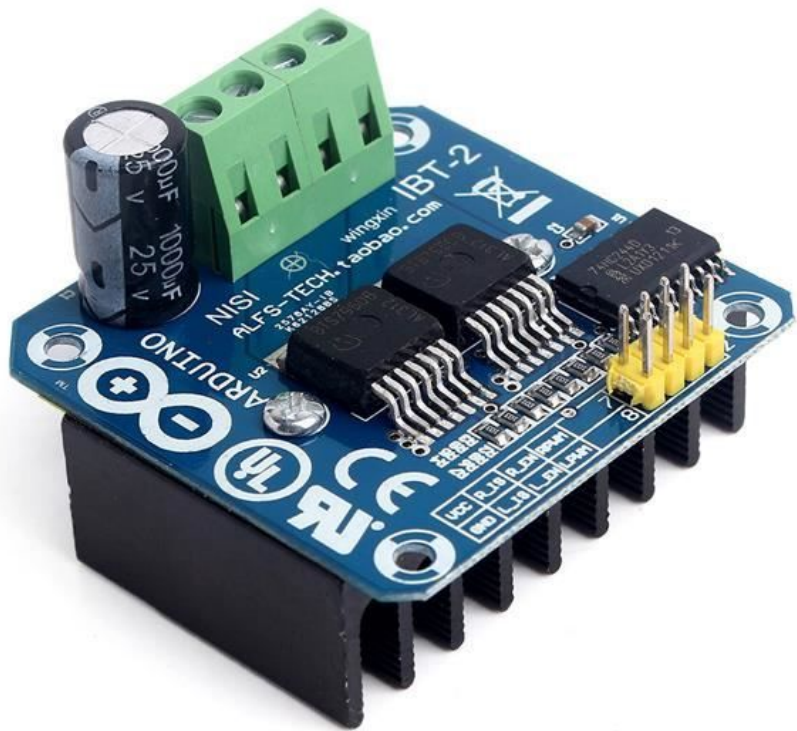


BTS7960 MOTOR DRIVER WITH DC MOTOR

At the heart of the module is the big, black chip with chunky heat sink is an L298N.



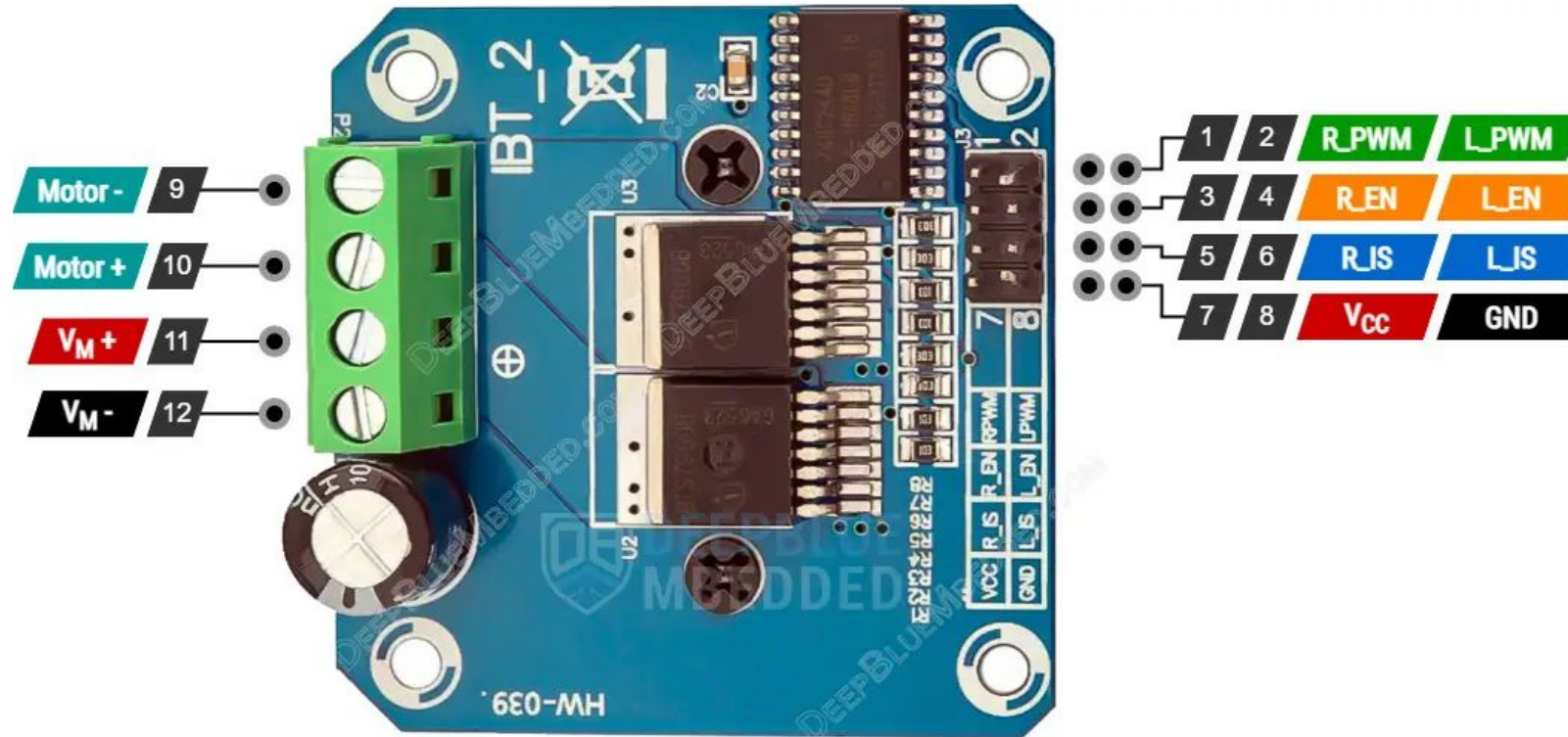
BTS7960 MOTOR DRIVER WITH DC MOTOR



BTS7960 MOTOR DRIVER WITH DC MOTOR



BTS7960 MOTOR DRIVER WITH DC MOTOR



BTS7960 Motor Driver **Pinout**

BTS7960 MOTOR DRIVER WITH DC MOTOR

BTS7960 Power Pins

V_M+ is the motor supply pin. It can accept any voltage within the 6v to 27v range. This needs to be connected to a separate supply other than the microcontroller's supply. Connecting it to the Arduino's 5v output will damage your Arduino board by over-driving its LDO or induce a lot of noise to the 5v line that feeds the microcontroller and potentially cause it to randomly reset.

$V_{CC}(5v)$ is the logic supply pin. This pin should be connected to the 5v to supply the internal logic circuits inside the BTS7960 motor driver IC.

GND is the ground pin. This pin should be connected to the ground of the Arduino board & the ground of the motor power supply (V_M).

BTS7960 MOTOR DRIVER WITH DC MOTOR

Motor Output Pins

Motor + & **Motor -** are the output pins of the H-Bridge in BTS7960 that connect to the Motor terminals.

Motor Enable Pins

R_EN & **L_EN** are the motor output enable pins for forward drive and reverse drive respectively.

Those pins are used for motor drive enable of both directions. You can hook them up to any output pins of the Arduino or just connect both of them to +5v to keep the bridge enabled and control the speed using the PWM pins only.

Motor PWM (Speed Control) Pins

R_PWM & **L_PWM** are the motor speed control inputs (PWM) for both the forward and reverse drives. You can control the speed and direction of the DC Motor using those two inputs only.

BTS7960 MOTOR DRIVER WITH DC MOTOR

x_IS Current-Sense & Diagnosis Pins

R_IS & **L_IS** are the current sense and diagnosis output pins for both the forward & reverse drives.

BTS7960 MOTOR DRIVER WITH DC MOTOR

```
// Motor A connections
int enA = 9 , in1 = 8 , in2 = 7;
// Motor B connections
int enB = 3 , in3 = 5 , in4 = 4;

void setup() {
    // Set all the motor control pins to outputs
    pinMode(enA, OUTPUT);
    pinMode(enB, OUTPUT);
    pinMode(in1, OUTPUT);
    pinMode(in2, OUTPUT);
    pinMode(in3, OUTPUT);
    pinMode(in4, OUTPUT);
    // Turn off motors - Initial state
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
    digitalWrite(in3, LOW);
    digitalWrite(in4, LOW);
}
```

Strategies

strategy

strategy

strategy

**YOU ARE READY TO MAKE YOUR OWN
SUMO ROBOT NOW**





RESOURCES



And many more resources either websites, communities or Youtube channels.

“LEARNING HAS NO END”

